



Documentation

Table of Contents

1 General Information

I Inspectrum.Suite Software

2 Software Licensing

3 Software Installation

- 3.1 System Requirements
- 3.2 Installation procedure

4 Software Overview

- 4.1 Toolbar
- 4.2 Workspace Explorer
- 4.3 Main Area

5 Basic device setup

- 5.1 Connect a Battery

6 Start Measurement

- 6.1 Single Measurement
- 6.2 Continuous Measurement
- 6.3 Sweep Measurement (Pro version only)
- 6.4 Burst Measurement (Pro version only)

7 Results and Live Data

- 7.1 Measurement Results
 - 7.1.1 Managing plot traces
- 7.2 Spectrum Compensation
- 7.3 Battery Monitor
 - 7.3.1 Charge/discharge process

8 File Import and Export

- 8.1 Export measurement results
 - 8.1.1 Native format export
 - 8.1.2 Static PNG plot export
 - 8.1.3 Interactive HTML plot export
 - 8.1.4 CSV conversion
 - 8.1.5 CSV Import
 - 8.1.6 Device configuration export
 - 8.1.7 Measurement results import
 - 8.1.8 Device configuration import
 - 8.1.8.1 From configuration file (.icf)

- 8.1.8.2 From the measurement results file (.irf)

9 Multiplexer Control

10 Device Configuration

- 10.1 Excitation
 - 10.1.1 Excitation Signal Editor
 - 10.1.2 Signal optimization
 - 10.1.3 Excitation Signal Wizard (Pro version only)
- 10.2 Limits
- 10.3 Multiplexer
 - 10.3.1 Multiplexer Sequence
- 10.4 CAN Interface
 - 10.4.1 CAN Matrix Editor
- 10.5 Miscellaneous
- 10.6 Device Information

11 Preferences

- 11.1 Workspace
- 11.2 Export
 - 11.2.1 Battery Names Setup
- 11.3 Display
- 11.4 Miscellaneous

12 CAN Interface

I.I Miscellaneous

13 Firmware Update

14 Plugins (Pro version only)

15 Changelog

- 15.1 Version 2.6 (April 2024)
 - 15.1.1 New features
 - 15.1.2 Bugfixes

16 Troubleshooting

- 16.1 Error Codes

II Device Automation Plugin

17 Introduction

18 Installation

19 Overview

- 19.1 Context Menu
- 19.2 Control panel

20 Schedule Creation

- 20.1 Instructions
- 20.2 Description of the instruction set

21 Scripting

- 21.0.1 Introduction
- 21.0.2 Loading Existing Scripts
- 21.0.3 Testing Python Scripts Prior to Deployment
- 21.1 Executing Scripts
 - 21.1.1 Configuring Script Parameters
 - 21.1.2 Using Measurement Results
 - 21.1.3 Initiating Script Execution
 - 21.1.4 Troubleshooting

22 Preferences

III Model Generator Plugin

23 Overview

24 Installation

25 Model Generation

- 25.1 Input Data
 - 25.1.1 Average Groups:
- 25.2 Validation
- 25.3 Model Structure
 - 25.3.1 Fitting Results
 - 25.3.2 Parameter Adjustment
- 25.4 Exporting Results
- 25.5 Importing Results

26 Preferences

Version 2.6

1 General Information

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I. Inspectrum.Suite Software

2 Software Licensing

The Inspectrum Suite is available in two versions: Lite and Professional.

Some software functionalities are only available in the Professional version.

Feature	Lite	Professional
Single-sine/Multi-sine Measurement	✓	✓
Single-sine/Multi-sine Sweep Measurement	✗	✓
Burst Measurement	✗	✓
Excitation Signal Wizard	✗	✓
Expandable via Plug-Ins	✗	✓
Supports Multiplexer & CAN Interface	✓	✓
Export Formats	Native (xml), png	Native (xml), png, html, csv

With your device, you will get a license dongle for the software version you bought.

This dongle must always be connected to your PC to properly start the software.

3 Software Installation

3.1 System Requirements

OS: Windows 10 64-bit (recommended) or Windows 11

Processor: 64-bit processor, 1 Gigahertz (GHz) or faster

RAM: 2 Gigabytes (GB)

Hard Disk Space: At least 1 GB

3.2 Installation procedure

To simplify the installation of the Inspectrum software, an installation wizard "setup.exe" is provided in the software folder:

 InspectrumApplicationSetup	13.10.2021 10:16	Windows Installer-Pa...	89.591 KB
 setup	13.10.2021 10:15	Anwendung	521 KB
 ftdidriver_x64	13.10.2021 10:31	Dateiordner	

Figure 1: Content of installer folder.

After running this file, the wizard will ask for permission to install the required FTDI D2xx COM driver. Please confirm the installation of the software package and follow the upcoming instructions to finish the driver installation.

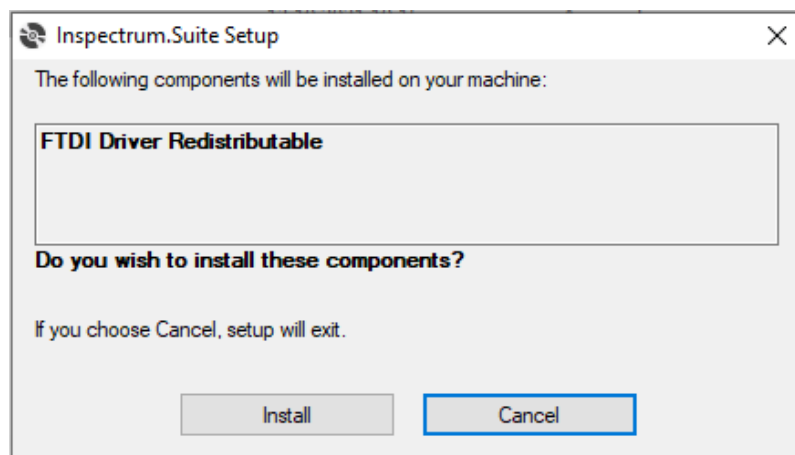


Figure 2: FTDI Driver Installation.

The installation wizard will now ask to install the Inspectrum.Suite software. After proceeding, the computer might ask for administrator rights. These must be granted to continue with the installation.

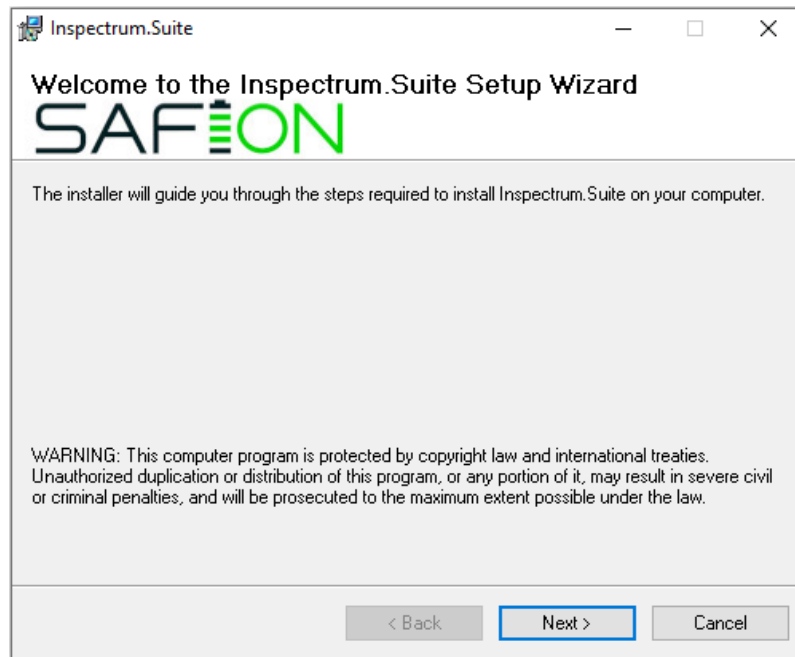


Figure 3: Installation Wizard

As soon as the setup wizard is finished, a system restart might be necessary. Once the installation is done, the Inspectrum.Suite software can either be found in the Windows start menu in the "Safion" folder, or by searching for "Inspectrum.Suite" in the Windows search bar.

4 Software Overview

After starting the "Inspectrum.Suite" software, the user interface will open. In case an ultra-high-definition monitor (4K) is used, it might be necessary to modify the display's scaling in your system settings to ensure a proper visualization of the control window.

The window consists of three main elements:

1. Toolbar with control elements
2. Workspace Explorer with access to your local files
3. Main Area (Measurement Results / Battery Monitor)

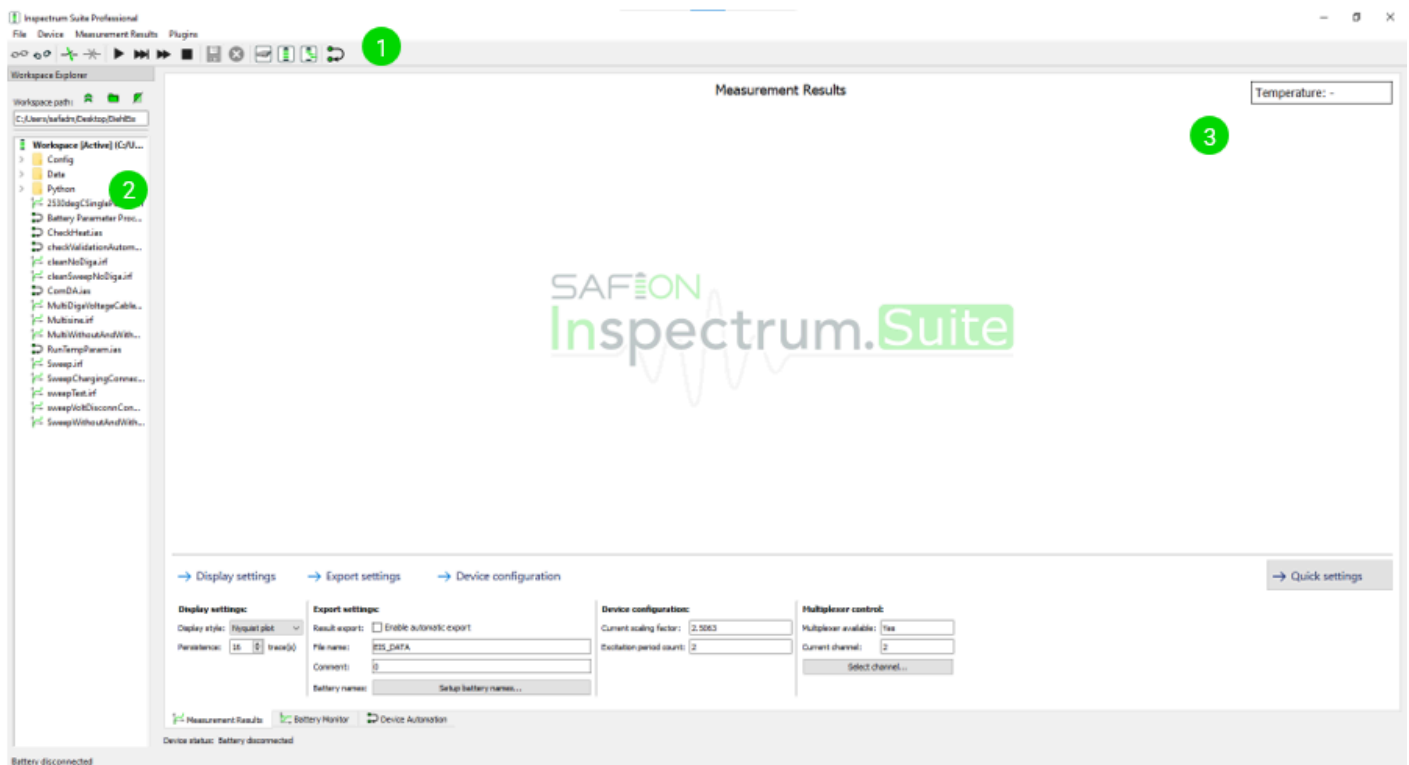


Figure 4: Main Window of Inspectrum.Suite.

4.1 Toolbar

The toolbar at the top left allows quick access to the most essential functions of the Inspectrum.Suite. An overview of each function is given in the table below.

1. Connect to device
2. Disconnect from device
3. Connect battery
4. Disconnect battery
5. Start single measurement
6. Start sweep measurement
7. Start continuous measurement
8. Stop measurement
9. Export spectrum
10. Clear
11. Device configuration
12. Preferences
13. Plugin manager



Figure 5: Inspectrum.Suite Toolbar.

4.2 Workspace Explorer

1. Directory up
2. Browse directory
3. File filter
4. Tree view

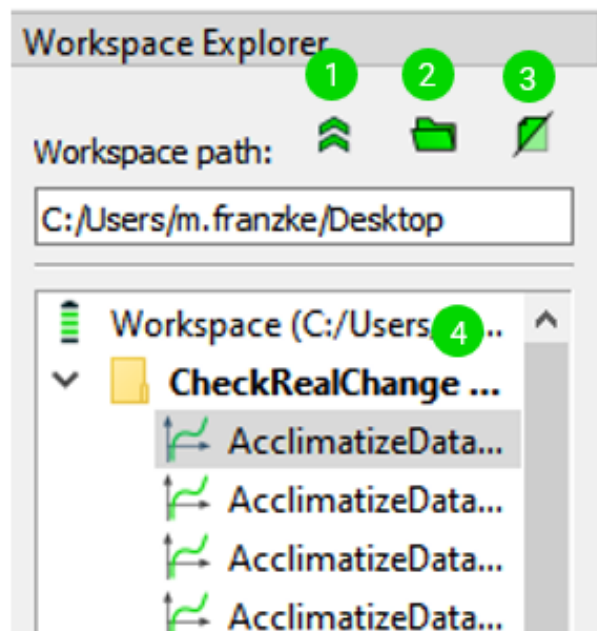


Figure 6: Workspace Explorer.

The Workspace Explorer on the left shows a tree view of the selected working path for importing and exporting measurement data. Here you can manually change or create new folders for your measurements. It allows for an easy organization of different data sets and lets you open and combine them via drag and drop. The active path is highlighted with bold letters.

The file filter allows you to select which file types should be displayed, helping to tidy up the view by showing only the chosen file types. You can set the Workspace path and active path by clicking on the *Preferences* button (12) in the toolbar or by right-clicking on a selected folder in the Workspace Explorer and selecting the 'set workspace path' option. For further explanation please go to [the preferences chapter](#).

4.3 Main Area

The Main Area can show two different kinds of information: Measurement result spectra and battery live data (Battery Monitor). You can switch between the two by clicking on the respective tab at the bottom left of the screen.

In the Measurement Results tab, the graphical overview of your measurement data will be shown. The quick settings tab at the bottom right allows for quick customization of the measurement process. The Display Settings, Export Settings and Device Configuration windows can also be accessed here for more customization options.

The Battery Monitor tab displays live data of the connected battery's voltage, current and temperature and allows one to charge or discharge the battery to a specific voltage level.

A detailed explanation of all the customization options for the display can be found in [the preferences chapter](#).

5 Basic device setup

Note

For detailed information on how to set up an **Inspectrum.ES** or **Inspectrum.HD** device and optionally connect a Multiplexer refer to the respective device documentation.

After the device has been powered up and has been linked to the PC via the provided USB cable, the device needs to be connected to the software. This is done by clicking on the *Connect to device* button in the toolbar (see chapter [Overview](#)). All communication interfaces and settings saved on the device will be loaded.



The Inspectrum.Suite software can handle only one device at a time. If multiple devices are connected during a connection attempt, the desired device has to be chosen in the Device Selection window.

Default configuration

The table below shows the default device settings when setting up your first measurement (if you upgrade from an older software version, some settings may vary). Depending on your battery type, some settings may need to be reconfigured for optimal results. To learn how to change the default device configuration, please refer to chapter [Device Configuration](#)

Parameter	Default setting
Over-current limit [A]	2.5
Over voltage [V]	4.2
Under-voltage [V]	2.5
Tolerate fault duration [ms]	1
Current scaling factor	1
Adaptive current scaling	Off
Target voltage amplitude for adaptive current scaling [mV]	18
Excitation period count	3
Resulting excitation duration [s]	2
Measurement enhancements	On
Continuous measurement period [s]	10

If you are using Inspectrum with the multiplexer and channel zero is selected when connecting the device, a pop-up window will ask you to select another channel. The channel can also be chosen in the quick settings section. Further information can be found in chapter 9.

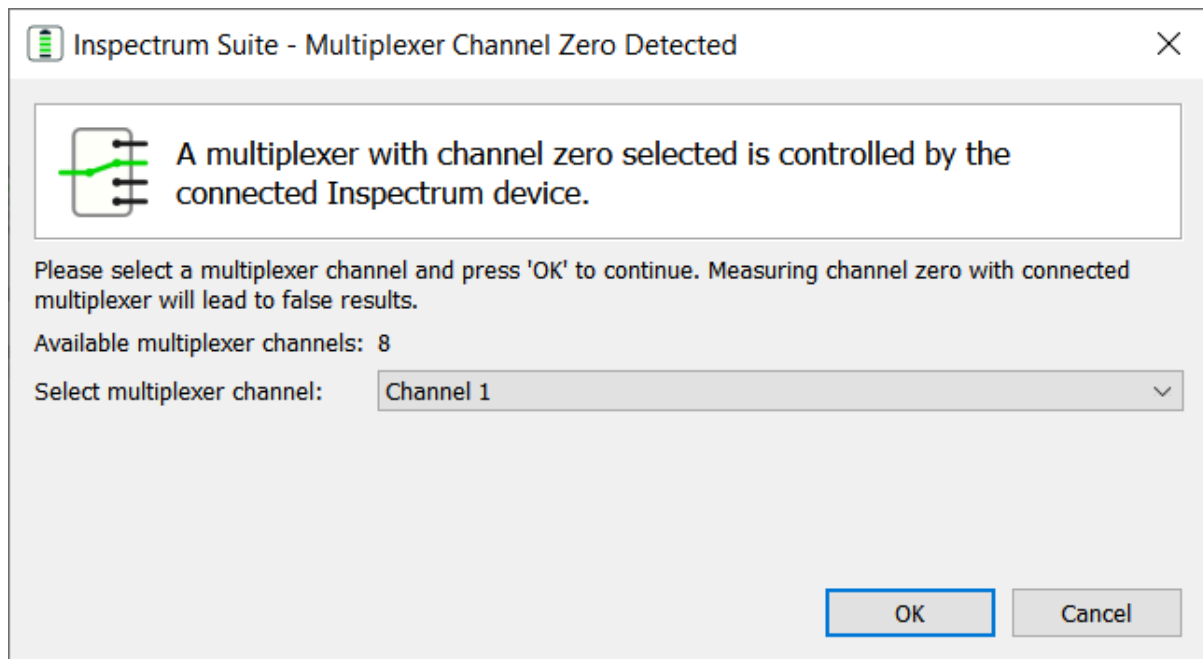


Figure 7: Multiplexer channel selection dialog.

By default, your measurement results will be automatically saved in the current active path. If you wish to disable the automatic export or change the file destination, click on the *Preferences* button in the toolbar and move to the *Export* tab. For a detailed explanation of these preferences, see chapter [Preferences](#).

5.1 Connect a Battery

Before activating the connection to the battery in the software, it needs to be physically connected to the device. Please make sure that the current, voltage and temperature Measurement cables are connected to the battery properly.

Connect the battery to the device by clicking the *Connect battery* button in the toolbar.



The device's status at the bottom left corner of the screen changes to "\"Battery connected\"" and the green "\"Connected\"" LED on the device lights up. You are now ready to take your first measurement.

6 Start Measurement

6.1 Single Measurement

Click on the *Start single measurement* button in the toolbar to initiate a single measurement.



The yellow *Active* LED on the device turns on. As soon as the measurement is finished, the LED turns off, and the spectrum appears in the Measurement Results tab of the Main Area.

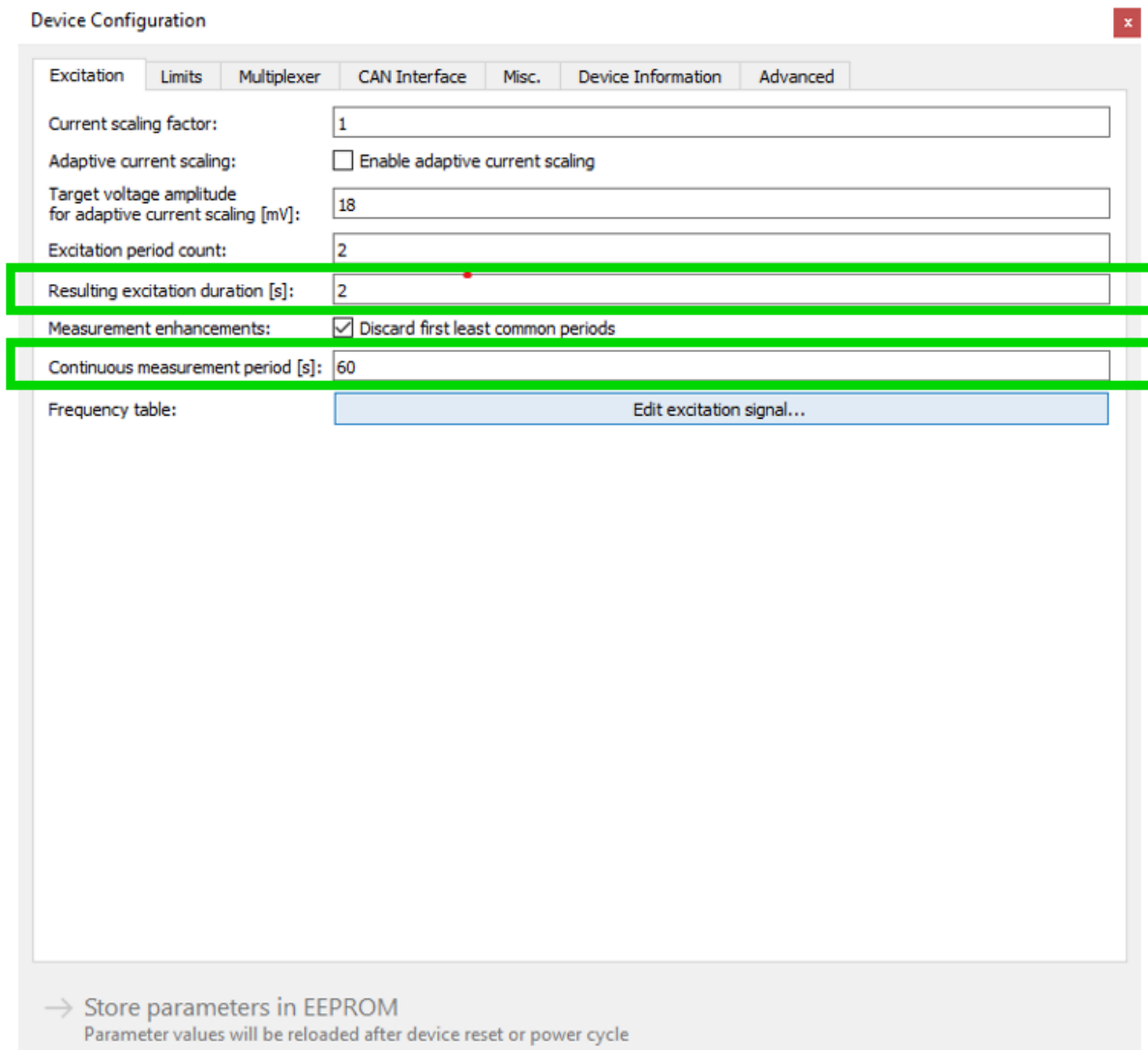
6.2 Continuous Measurement

The continuous measurement mode offers a possibility to gather electrochemical impedance data over a long period. This mode can be helpful in understanding a system's behavior in changing environmental conditions. The Inspectrum device will repeatedly perform multi-sine measurements until the desired number of repetitions is reached, or it is manually terminated.

Before starting this measurement, some configurations must first be set. Click the *Device Configuration* button in the toolbar, move to the *Excitation* tab, and set the parameter *Continuous measurement period* to a value matching the desired measurement interval. It can also be set to zero to start the next measurement as soon as possible.

Important information

- The value displayed in the *Resulting excitation duration* field limits the minimum achievable measurement period. This field is calculated from the lowest frequency point times the *Excitation period count*.
- If a multiplexer is connected and the channel sequencer is enabled (chapter 9), the *Resulting excitation duration* must be multiplied by the number of channel sequence steps.



Device Configuration

Excitation Limits Multiplexer CAN Interface Misc. Device Information Advanced

Current scaling factor: 1

Adaptive current scaling: ☐ Enable adaptive current scaling

Target voltage amplitude for adaptive current scaling [mV]: 18

Excitation period count: 2

Resulting excitation duration [s]: 2

Measurement enhancements: ☒ Discard first least common periods

Continuous measurement period [s]: 60

Frequency table: Edit excitation signal...

→ Store parameters in EEPROM
Parameter values will be reloaded after device reset or power cycle

Figure 8: Settings for continuous measurement mode.

Afterwards, click the *Start continuous measurement* button in the toolbar to initiate a continuous measurement.



In the pop-up window, set the number of repetitions for the measurement and press *OK* to start the measurement.

The measurement results of each repetition will automatically be shown in the Main Area. To manually stop the measurement, click the *Stop measurement* button in the toolbar.



6.3 Sweep Measurement (Pro version only)

The Sweep measurement method is used whenever it is not desired to measure all frequencies from the frequency table simultaneously. This Sweep measurement can either be done by measuring one frequency at a time with the Single sine sweep option or by measuring packets of frequencies with the Multi sine sweep option. Click on the *Start sweep measurement* button in the toolbar to configure your sweep measurement.



As a result, a window will open that allows you to set the excitation frequencies and amplitudes for your measurement.

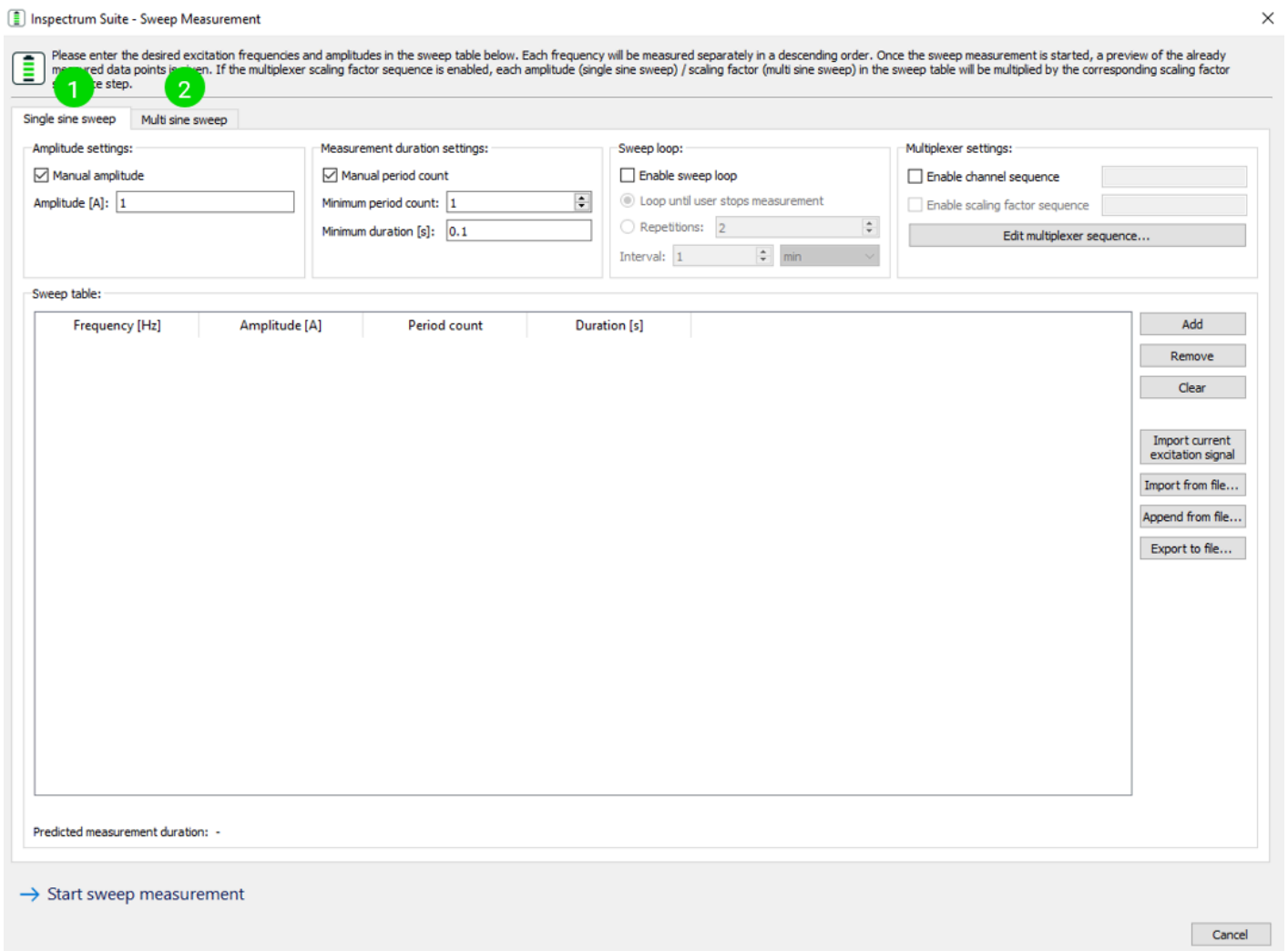


Figure 9: Sweep measurement dialog.

You can initiate a *Single sine sweep* (1) or a *Multi sine sweep* (2) if you want to measure multiple frequency points.

The excitation signal for these measurement modes can be directly imported from the current settings or a file. Editing these signals and adding single frequencies is also possible in this window. In the *Single sine sweep* option the amplitude and periods for each frequency and minimum duration can either be selected manually or set to the same value for all frequencies. For the *Multi sine sweep*, multiple excitation signals can be specified. These must be selected in such a way, that no frequencies overlap.

For both the *Single sine sweep* and the *Multi sine sweep* options, you can enable a sweep loop setting that will either loop infinitely until you stop the measurement, or you can manually set the number of repetitions and the interval duration.

In case a Multiplexer is connected to the device, enable channel sequencing and scaling factor sequencing for each channel here. Click on *Edit multiplexer sequence* to specify the sequence length of your multiplexer measurement and adjust the channel number and scaling factor for each respective channel, if necessary. Please read chapters 10.1 and 10.3 for a detailed description.

Click on the *Start sweep measurement* button to initiate the sweep measurement. A preview of the data points will be shown in the Measurement Results tab of the Main Area.

6.4 Burst Measurement (Pro version only)

The burst measurement mode focuses on performing multiple measurements in rapid succession. This mode uses a faster voltage measurement and skips the temperature measurement. Additionally, the results are stored and will only be displayed after completing the measurement.

Click on *Device* → *Start burst measurement* to initiate a burst measurement. Note that the burst measurement will clear the current measurement results and import burst results after the measurement has been finished.

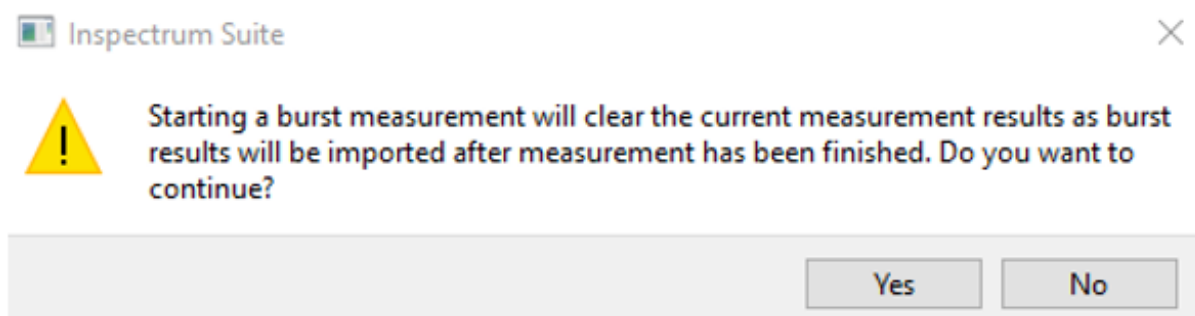


Figure 10: Dialog warning the user that currently displayed measurement results will be cleared after burst mode.

During the execution of the burst measurement, information like the number of measurements performed, the frequency of measurements and the number of clipped measurements will be displayed in a new window.

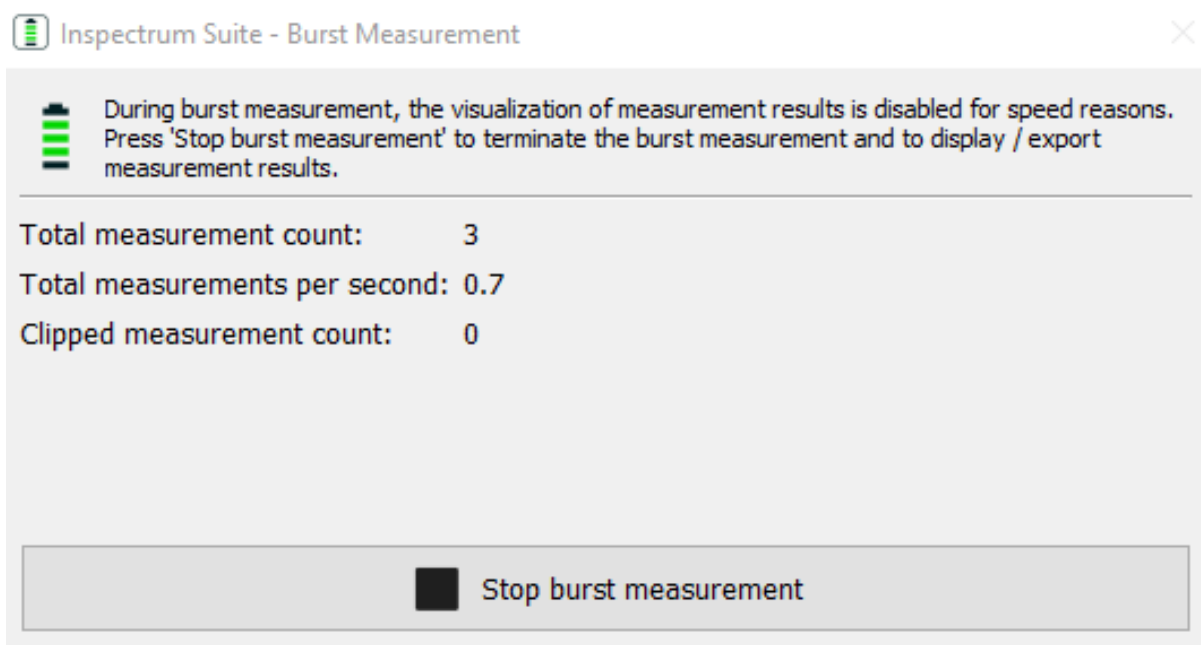


Figure 11: Popup dialog shown during burst measurement.

After the measurement is finished, the results will be shown in the Main Area. To manually stop the measurement, click on the *Stop burst measurement* button at the bottom of the window.

7 Results and Live Data

7.1 Measurement Results

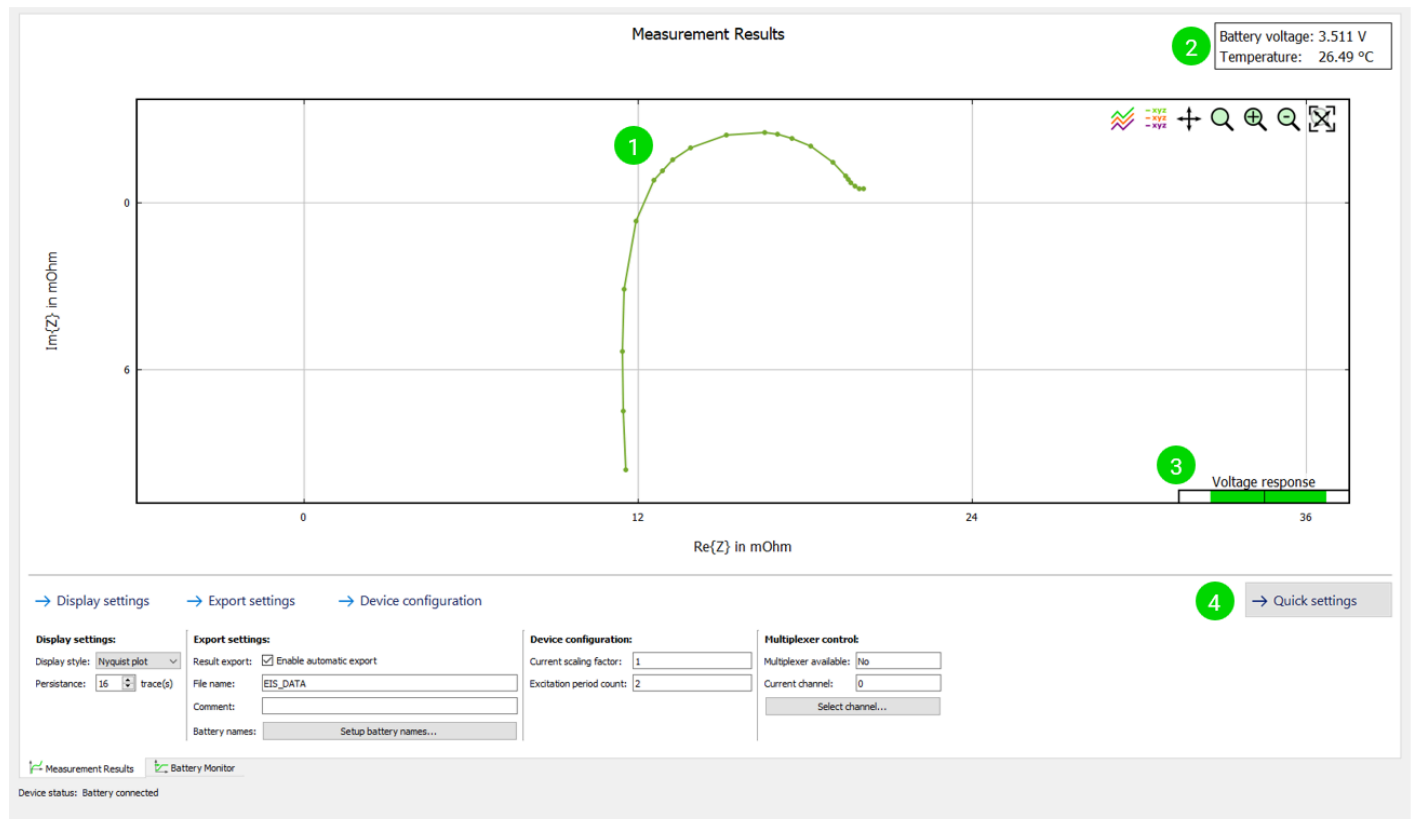


Figure 12: Measurement results view.

The results of your measured impedance data (1) will automatically be shown in the Measurement Results tab of the Main Area.

The top right corner shows a quick overview of the battery voltage and temperature (2).

The voltage response of the battery (3) is displayed in the bottom right corner showing the voltage deflection of the cell relative to the measurement range of the analog frontend. It can be used as an indication for the measurement resolution and a guidance for tuning the excitation current, for example by adjusting the *Signal scaling factor*.

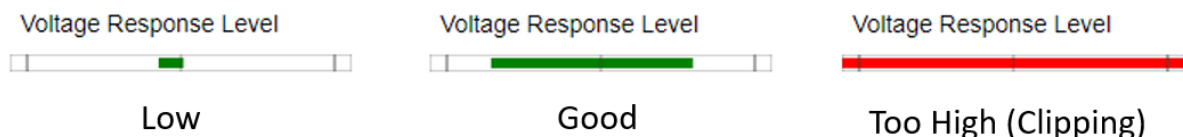


Figure 13: Voltage response level indicator.

The most commonly used settings for the display, data export, device configuration and multiplexer control are available in the *Quick settings* (4) underneath the plot. Here, you can switch from the default Nyquist Plot to a Bode Plot or set the number of traces to be plotted. Furthermore, you can set up battery names, change file names, set the current scaling factor and the excitation period count, or switch the Multiplexer channel (if a Multiplexer is connected to the device).

Clicking on the respective settings button will lead you to further customization options for [device configuration](#) and [software preferences](#).

7.1.1 Managing plot traces



Figure 14: Plot traces toolbar.

By clicking the "Select traces" icon within the plot toolbar on the upper right side of the plot window, the trace editor dialog can be opened:

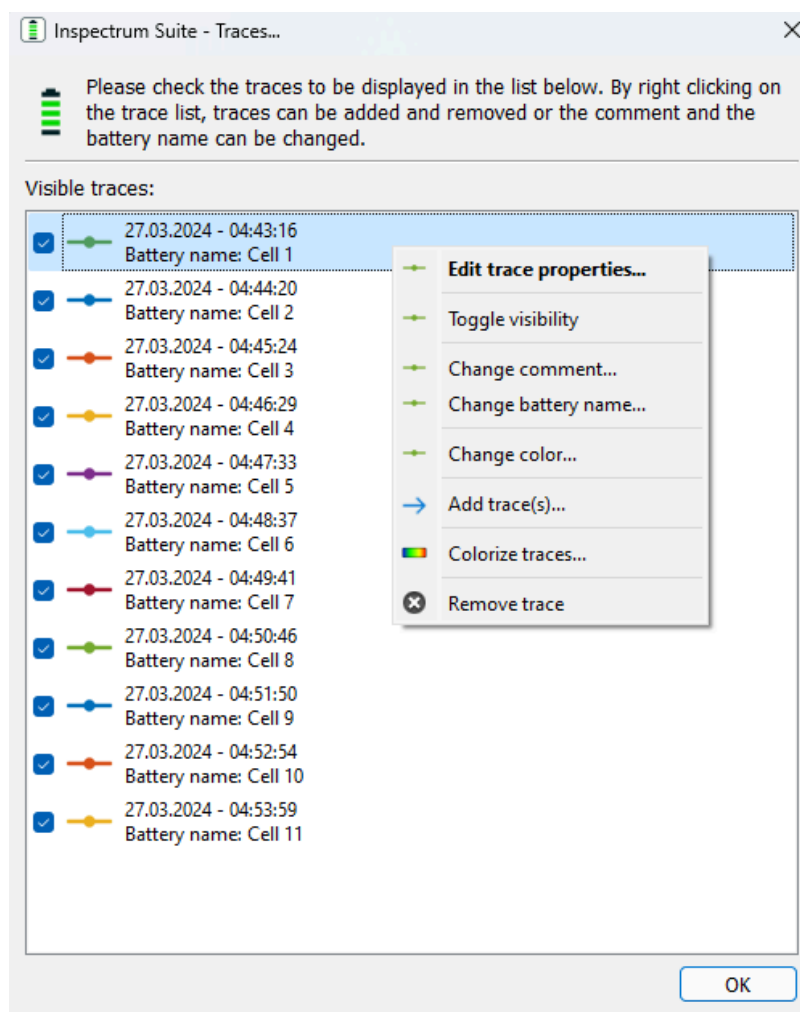


Figure 15: Trace editor dialog.

By clicking the checkboxes of the individual traces on the left side, their visibility within the plot window can be toggled. A right-click on the traces opens the context menu, which allows editing further properties, like color, battery name and comment.

Using the Shift key and left click, multiple traces can be selected:

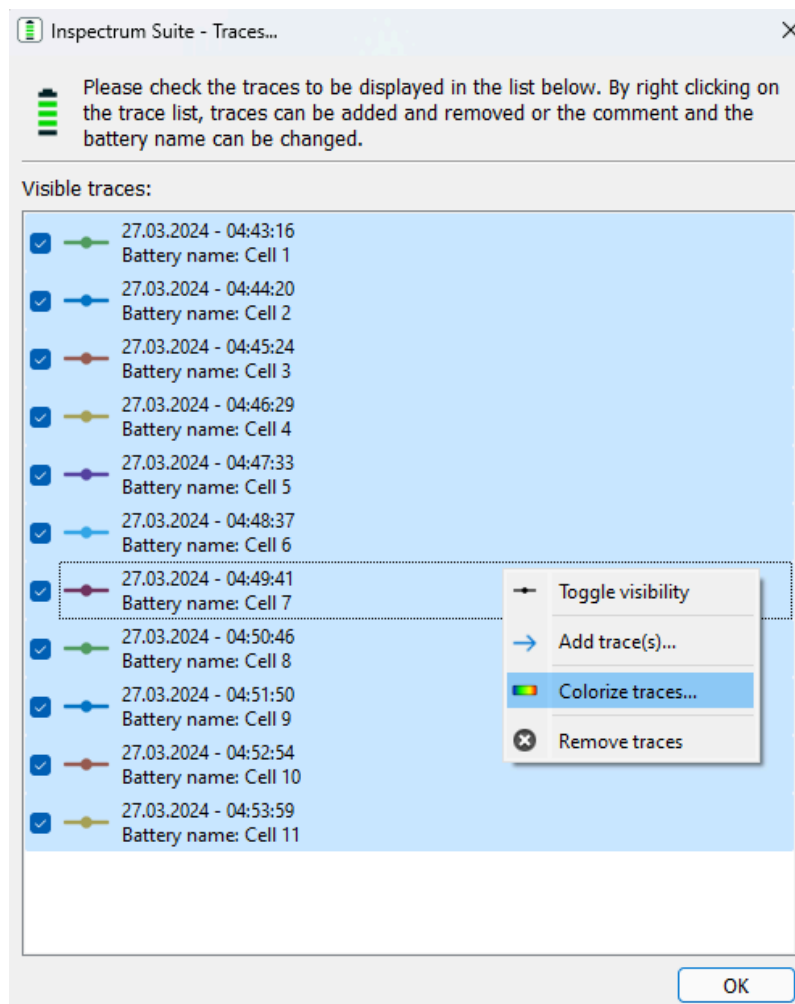


Figure 16: Multi-Select in the Trace editor dialog.

Using a right click on multiple selected traces, the context menu provides a reduced set of options.

The **Colorize Traces** function provides a very useful functionality for certain types of measurements: It allows automatically setting a color gradient for the selected traces, based on either the voltage or temperature. This is useful to visualize impedance data obtained under varying conditions:

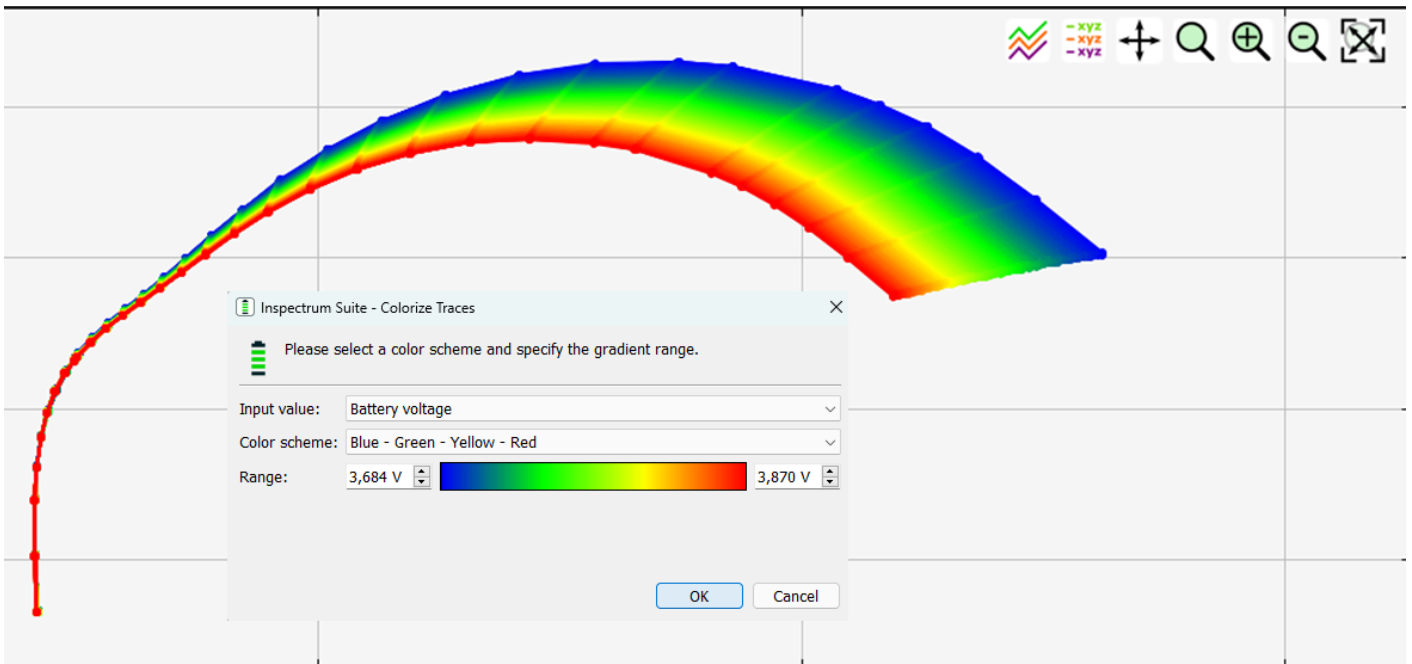


Figure 17: Colorize traces function.

Within the dialog, changing the input value will automatically predefine a suitable range, using the maximum and minimum value of temperature or voltage respectively. However, this limits can also be adjusted manually. If the input value for the individual trace lies outside the defined range, the color of the maximum or minimum value will be assigned to it.

7.2 Spectrum Compensation

The spectrum compensation function allows to define a *Reference Spectrum*, which is subtracted from the actual measurement, using the following formula.

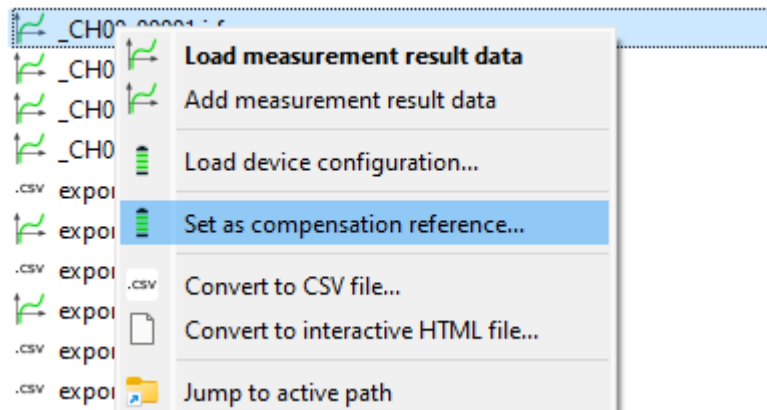
$$\underline{Z}_{comp}(j\omega) = \underline{Z}_{meas}(j\omega) - \underline{Z}_{ref}(j\omega)$$

This can be used for two application areas:

1. Eliminating the influence of wire routing and the cell holder or jig, by performing a reference measurement under short-circuit conditions.
2. Viewing *Differential EIS* (sometimes also called *Delta-EIS*) results, when comparing compensated spectra to measurements conducted on a different cell specimen or under different conditions.

The first step to use the compensation is to set a reference spectrum, which is applied to further measurements or loaded measurement data. This can be done in different ways:

- Using the context menu from within the workspace explorer:



- From the quick settings widget:

Spectrum Compensation:

Apply to: Disabled ▾

Compensation Source: C:/Users/h.zappen/InspectrumSuite/demo_compe ▾

Import...

Mode: Both ▾

- From within the **Display settings** dialog

Within the *Compensation Source* drop-down menu in the quick settings widget, the most recently used compensation source files can also be loaded. Two more options are available: - The *Apply to* drop-down menu allows to either apply the compensation to be applied to existing as well as new measurement data automatically, or limit it to new measurements or newly loaded data, or disable it completely. - The *Mode* drop-down allows configuring if the compensation is applied to the real part only, imaginary part only, or both.

A more fine grained control is also available for individual traces: Within the **trace editor**, right-click an individual trace and select *Edit trace properties...* By clicking the corresponding checkboxes, you can enable or disable the compensation for the selected trace, or even load custom compensation data, which is only applied to this trace.

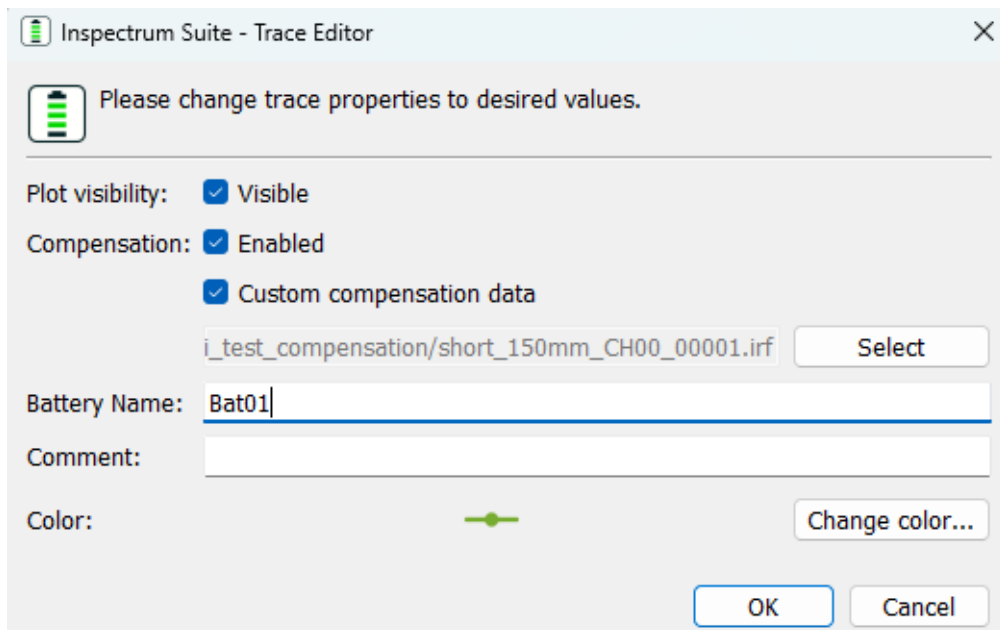


Figure 18: Use the trace editor to customize the compensation for individual traces.

7.3 Battery Monitor

To view the live data of your battery, please switch to the Battery Monitor tab at the bottom left of the Main Area. In the Battery Monitor window, the voltage and DC current output data of the connected battery will be displayed in real-time, depending on which sensors are attached.

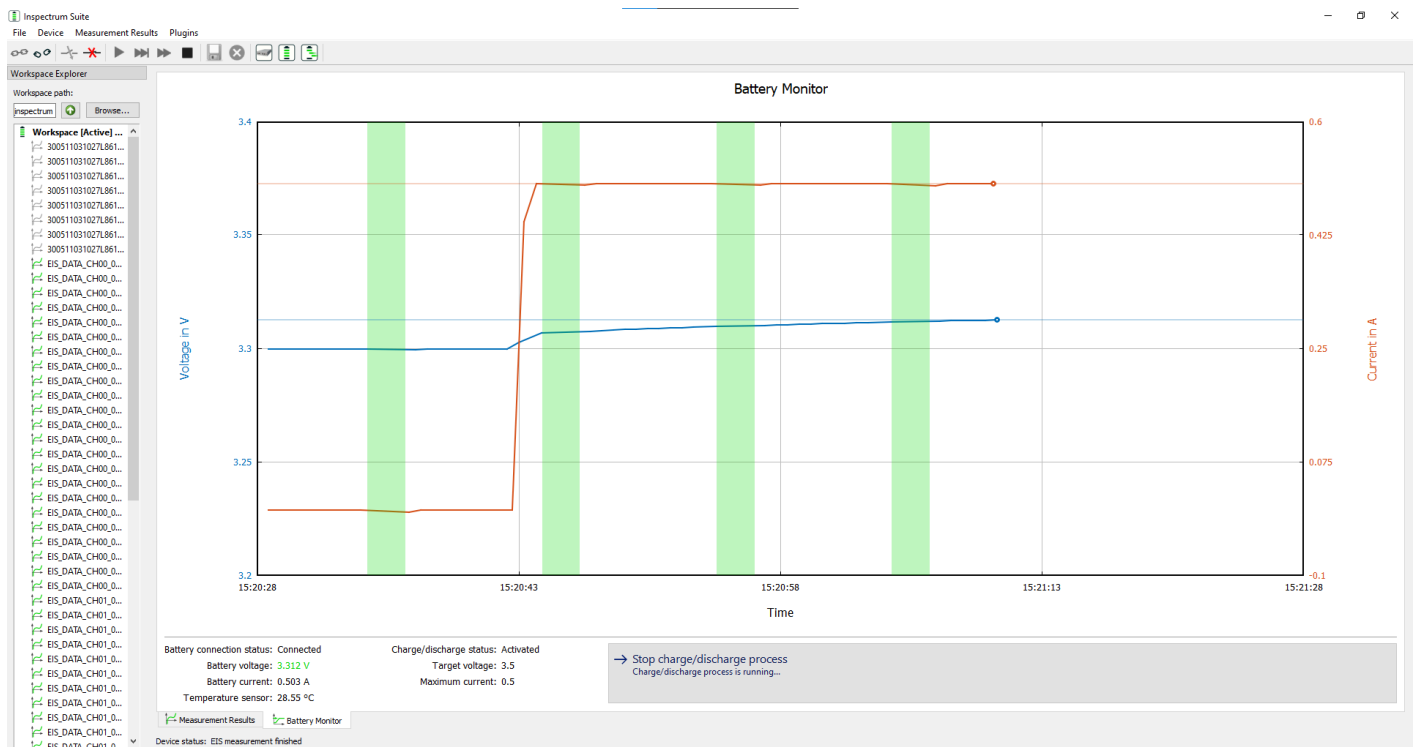
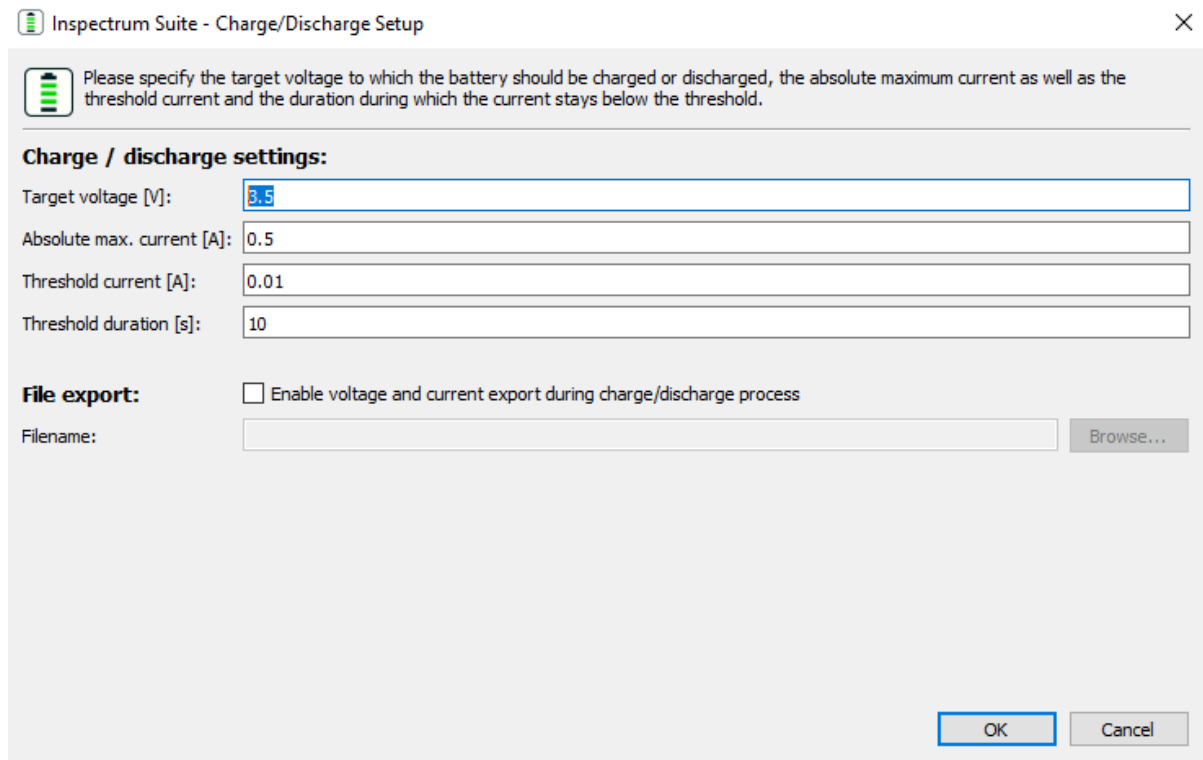


Figure 19: Battery monitor view.

7.3.1 Charge/discharge process

If the battery voltage of the connected cell is too low or too high, the Inspectrum.C can function as a charger or discharger to bring the cell to the desired measurement conditions. For this, click on *Start charge/discharge process* within the battery monitor view.



Inspectrum Suite - Charge/Discharge Setup

Please specify the target voltage to which the battery should be charged or discharged, the absolute maximum current as well as the threshold current and the duration during which the current stays below the threshold.

Charge / discharge settings:

Target voltage [V]: 3.5

Absolute max. current [A]: 0.5

Threshold current [A]: 0.01

Threshold duration [s]: 10

File export:

☐ Enable voltage and current export during charge/discharge process

Filename: Browse...

OK Cancel

Figure 20: Charge/Discharge process setup dialog.

Now, the battery target voltage and the maximum charge or discharge current must be configured. The Inspectrum.C can source or sink a DC current of 1 A. You can specify a threshold for the minimum current at which the charging or discharging process is stopped. This threshold must be exceeded for the time defined in the *Threshold duration* field to become valid.

Enable *File export* to have your voltage and current data exported into a CSV file after completing the charge/discharge process. For this, you must define a filename and export location.

When pressing *OK*, the charging/discharging process starts, and the current and voltage data of the connected cell will be displayed in the plot. Additionally, the *Measurement* LED at the front of the device will turn on. The process will stop automatically when the threshold condition is met and can be manually stopped by pressing *Stop charge/discharge process*.

Charging and Discharging of Batteries

Special care must be taken that the target voltage set in the software is within the limits of the cell type under test. If the cell is accidentally overcharged, there is a risk of electrolyte spillage and fire.



8 File Import and Export

8.1 Export measurement results

8.1.1 Native format export

By default, your measurement results are automatically exported to your workspace's active path after a measurement is finished. Alternatively, you can export the data manually by clicking the *Export* button in the toolbar or *File* → *Export* → *Export measurement results*. You can also export results by right-clicking on the plot.



The resulting *.irf* file contains all measurement data, as well as excitation settings and limits parameters of the used device configuration. It uses an XML format and can also be viewed using a text editor.

To adjust additional export settings, click on the *Preferences* button in the toolbar and move to the *Export* tab. A detailed explanation of the customization options is given in the [preferences chapter](#).

8.1.2 Static PNG plot export

To export the Measurement results graph as a static image, follow the steps to export measurement results and use *.png* as filetype.

8.1.3 Interactive HTML plot export

To export the currently displayed plot into an interactive HTML file, follow the steps to export measurement results and in the export dialog choose *.html* as filetype. Additionally, you can right-click on a file in the workspace explorer and choose *Convert to interactive HTML file....* When multiple files are selected you can choose between exporting them into one HTML file or creating a separate file for each.

8.1.4 CSV conversion

To generate a csv file from existing measurement files, select the files you want to export in the workspace explorer. Then right-click the selection and choose *Convert to CSV file....* This will open a dialog window where you can edit the CSV export settings.

Please select CSV file export options.

Selected files: 2

Result count: 2

Destination path:

Separator:

Decimal point:

End of line:

Export format:

Quotation marks: ☐ Wrap values in quotation marks

Figure 21: CSV export dialog.

Parameter	Description
Selected Files	Number of files selected to convert.
Result count	Number of impedance spectra that will be included.
Destination path	Path where the CSV file will be saved.
Separator	Choose between comma and semicolon as field delimiter.
Decimal point	Choose between comma or point as decimal separator for values.
End of line	Select which control character to use as end of line delimiter.
Export format	Select which data to include in the CSV file.
Quotation marks	Choose whether to include additional quotation marks around values.

8.1.5 CSV Import

Measurement result files in the *Digatron Battery Manager 4 (BM4)* CSV format can also be imported.

In the workspace explorer, make sure CSV files are visible by enabling the proper filter settings. Right click a CSV file and select *Import Measurement Data*.

A dialog window opens, which allows to select the CSV type (current only *Digatron Battery Manager*), and if the temperature data from the Inspectrum Device or Digatron cyclers should be used. If the import is successful, the measurement data shows up within the measurement results view, and can also be exported in the native format.

Please note that, due to not all metadata being available within the CSV files, not all settings values can be properly exported.

If the import does not work, please contact Safion support for an example file with the proper format.

8.1.6 Device configuration export

To export the device configuration, click on *File → Export → Export device configuration*. A new window will open, where you can define which configuration categories should be included in the exported .icf file.

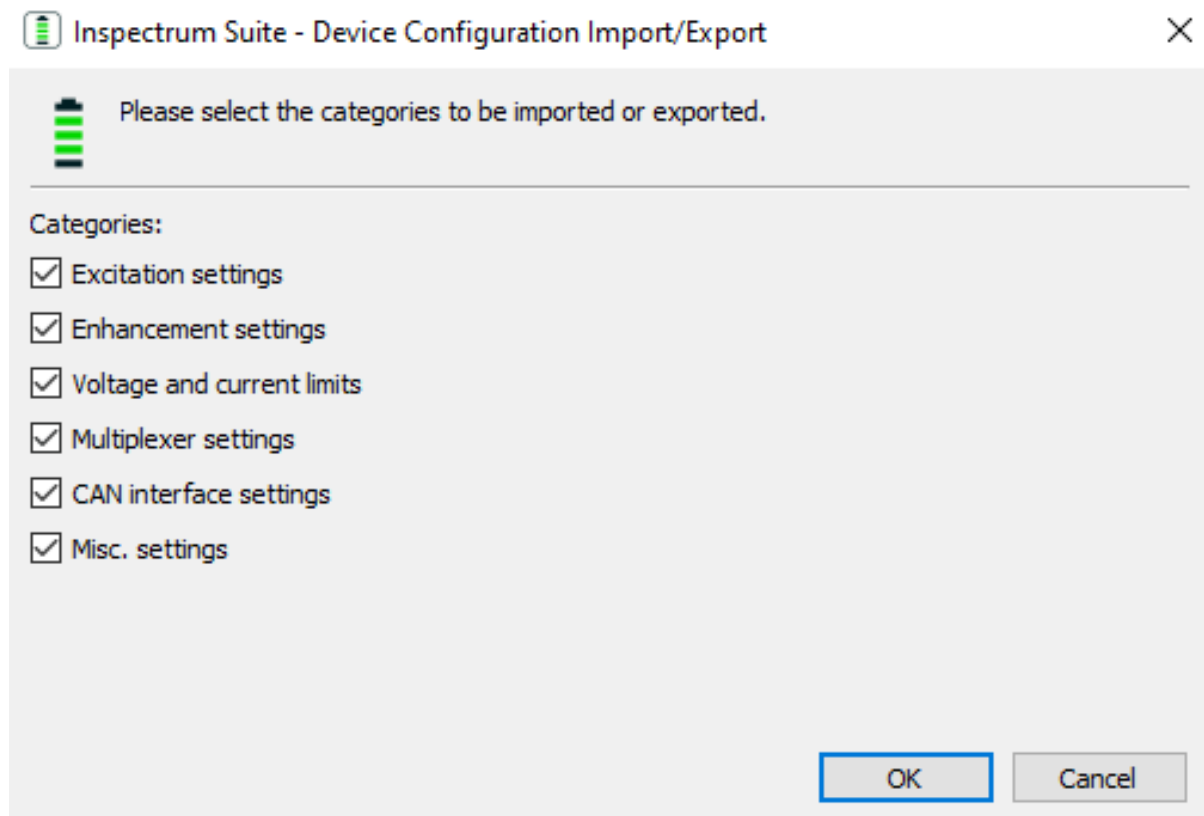


Figure 22: Device configuration export dialog.

8.1.7 Measurement results import

You can import measurement data of prior measured cells by clicking on *File → Import → Import measurement results*. Alternatively, you can drag and drop files directly from the Workspace Explorer or your Windows Explorer into the Measurement Results tab. Mark multiple files for import by holding CTRL or SHIFT.

8.1.8 Device configuration import

8.1.8.1 From configuration file (.icf)

If you want to use a previously exported device configuration, you can import the .icf file by clicking on *File* → *Import* → *Import device configuration*. After choosing the file, a similar dialog as during export is shown. Here you can define which configuration categories should be imported.

8.1.8.2 From the measurement results file (.irf)

You can also import the excitation and limits parameters for your device configuration from a measurement result file (.irf). For this, click on *File* → *Import* → *Device configuration*, then select .irf as the file type.

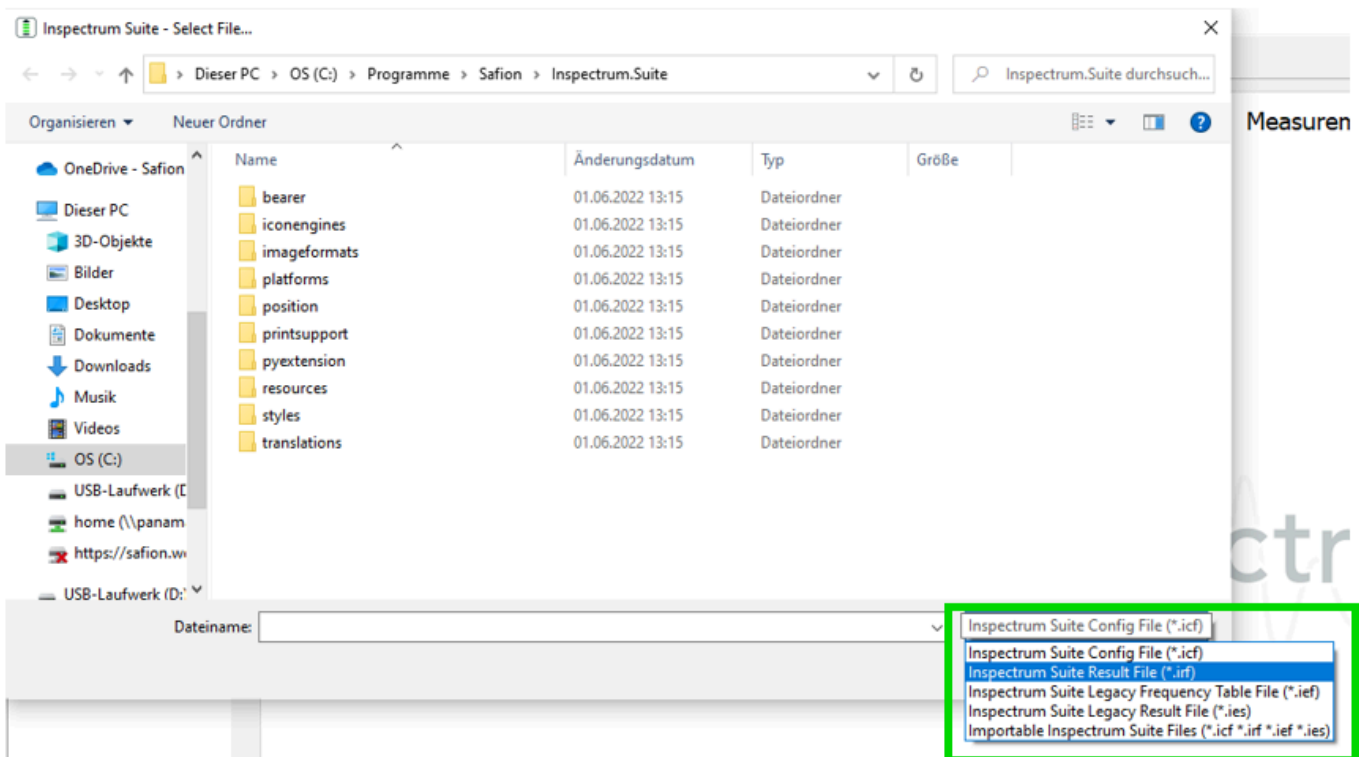


Figure 23: Import device configuration from result file.

Alternatively, you can right-click an .irf file in the Workspace Explorer and choose *Load device configuration*.

If the .irf file contains multiple spectra, you will have to choose the specific configuration you wish to import.

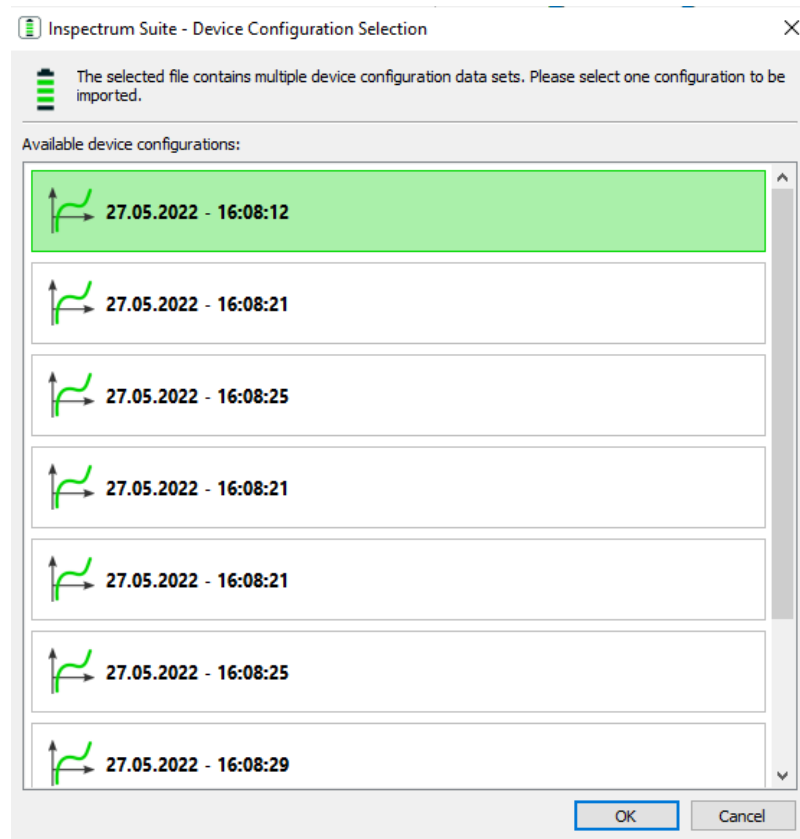


Figure 24: Device configuration selection dialog.

9 Multiplexer Control

Some devices, like the Inspectrum.ES can be extended with the MUX8 multiplexer. The Inspectrum.HD series comes with integrated multiplexers in different channel counts

If a multiplexer is detected, the *Multiplexer control* widget will show in the quick settings of the Main Area.

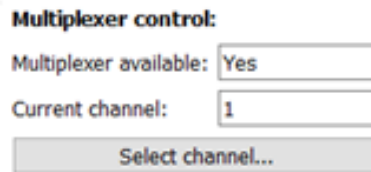


Figure 25: Multiplexer control widget.

The multiplexer can be controlled in two ways: manually with the Multiplexer control in the quick settings or with the automatic channel sequencer.

You can manually select a specific multiplexer channel by clicking on *Select channel* and choosing the desired channel. You can also deactivate the channel by clicking on *Select channel* and then on *Deselect channel*.

The other option is the automatic multiplexer sequencer. This function allows the multiplexer to automatically switch between each channel connected to a battery in ascending order. This is intended to be used in combination with the continuous measurement mode or the sweep measurement mode. Click on the *Device configuration* button in the toolbar and move to the *Multiplexer* tab to enable the automatic channel sequencing and the corresponding scaling factor or customize your multiplexer sequence. A detailed explanation of these configuration options can be found in the [device configuration chapter](#).

Please note that if the multiplexer is connected after the Inspectrum has been started, a restart is necessary for the Inspectrum to recognize it.

10 Device Configuration

To change your device's settings, click on the *Device configuration* button in the toolbar.



If you want to save your specified parameters and store them permanently on the device so that they are loaded again every time the device is powered on or restarted, click the *Store parameters in EEPROM* button.

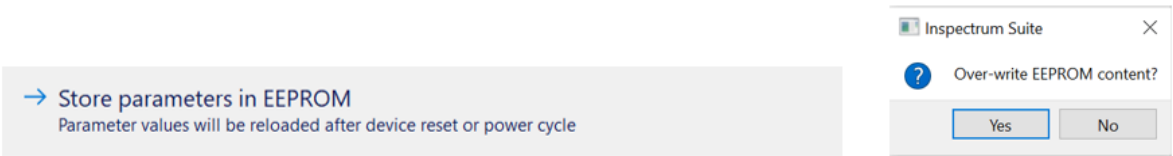


Figure 26: Write configuration data to device EEPROM.

10.1 Excitation

Device Configuration

ExcitationLimitsMultiplexerCAN InterfaceMisc.Device Information

Current scaling factor:

1

Adaptive current scaling:

☐ Enable adaptive current scaling

Target voltage amplitude for adaptive current scaling [mV]:

18

Period count (lowest frequency):

3

Resulting excitation duration [s]:

3

Measurement enhancements:

☒ Discard first least common periods

Voltage Drift Correction Mode:

Disabled

Continuous measurement period [s]:

10

Frequency table:

Edit excitation signal...

→ Store parameters in EEPROM
Parameter values will be reloaded after device reset or power cycle

Figure 27: Excitation configuration tab within device configuration.

Item	Description
Current scaling factor	Type in a factor for scaling the amplitude of the excitation signal. This factor is necessary if the cell's voltage response is too low or too high.
Enable adaptive current scaling	Enable this function to adjust the current scaling factor automatically after each measurement.
Target voltage amplitude	Type in a value for the setpoint cell voltage response, that should be reached during adaptive current scaling. After each measurement, the currently used scaling factor will be updated in the "Current scaling factor" field.
Excitation period	Define the number of excitation periods by entering an integer value equal or greater than one.

Item	Description
count	
Resulting excitation duration	This shows the resulting excitation period based on selected frequencies and the number of excitation periods.
Discard first least common periods	Enable this function to discard the first complete measurement period and only store subsequent periods for future use. Consequently, the "Excitation period count" needs to be greater than one so the measurement can be performed. This function can be beneficial in specific measurement setups.
Voltage Drift Correction Mode	Choose "Linear" or "Exponential" to enable drift correction. This is mostly useful for measurements during charge/discharge operation. It greatly reduces artifacts in the measurements due to voltage changes of the battery cell. For most cases, the "Linear" mode is recommended. The "Exponential" mode also removes exponential drift, for example during relaxation phases. It however requires that the period count is an odd number greater one, and that the "Discard first least common periods" option is active.
Continuous measurement period	Repetition rate for continuous measurement mode.
Frequency table	Click on "Edit excitation signal" to open the Excitation Signal Editor for creating, adjusting or loading a frequency table.

Important information

The excitation period count must be a multiple of the least common period count.

Within the software, the period of the signal is defined by the period of lowest frequency. Mind, that this does not hold, when there are frequencies in the signal whose period is not an integer divider of the period of the lowest frequency component. An example is given in the figure below:

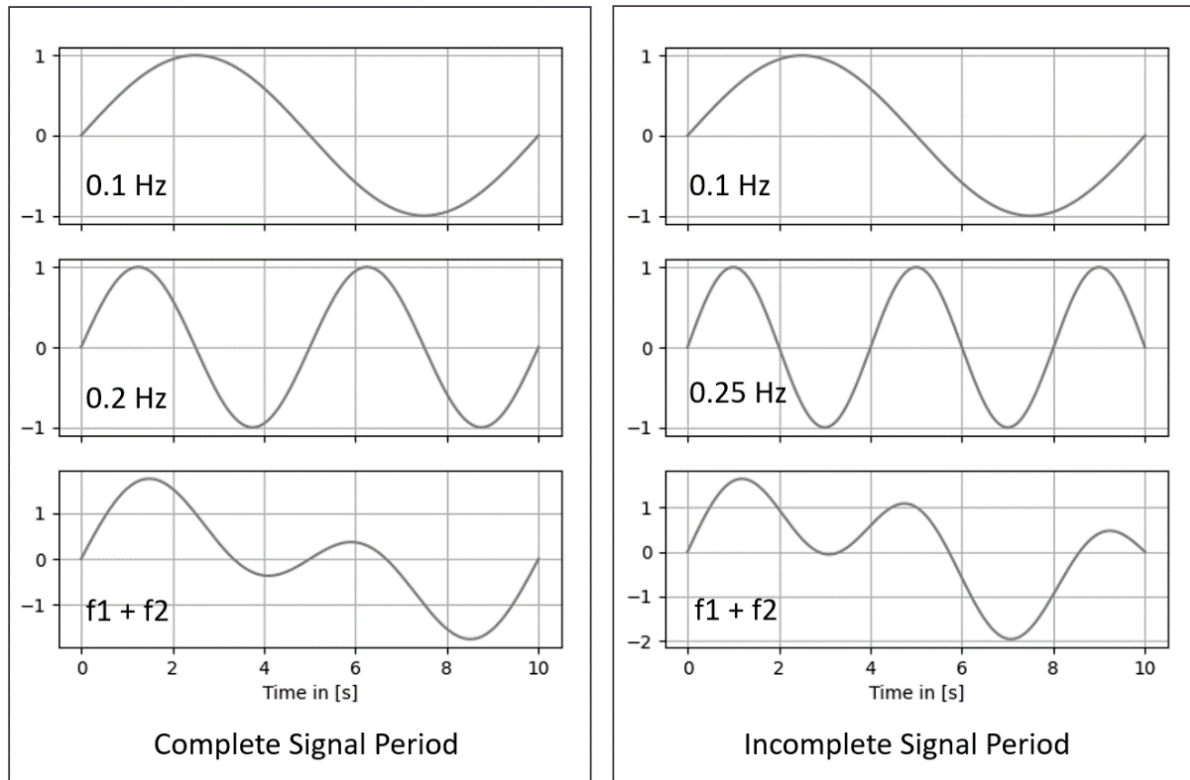


Figure 28: Sine signal period.

You must either choose suitable frequencies or select the number of excitation periods in such a way, that the displayed resulting excitation duration is a common multiple of all frequency component periods.

For the incomplete example given above, the number of excitation periods must be set to at least 2 in the software, resulting in a signal duration of 20 seconds.

Any other integer multiple of this base duration is also valid, for example, 4, 6, 8... periods.

Whenever you try to set an excitation period where this condition is not met, a message will be shown presenting an option to adjust the period automatically.

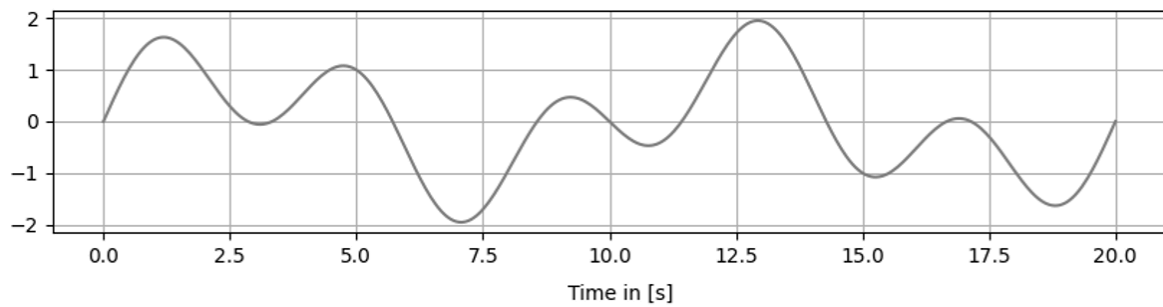


Figure 29: Multisine signal example.

10.1.1 Excitation Signal Editor

Inspectrium Suite - Excitation Signal Editor

Some computing intensive calculations may take a while depending on the excitation duration. The lower the lowest excitation frequency and the higher the period count, the longer the measurement duration and, therefore, the calculation time. The period count of the lowest frequency should be a multiple of the least common period count.

Current scaling:
Current scaling factor:

Excitation duration:
Period count (of the lowest frequency):
Resulting excitation duration [s]:

Frequency components:

	Frequency [Hz]	Amplitude [A]	Phase [p.u.]
1	1	0.06857	0.9258
2	2	0.06857	0.1016
3	3	0.06857	0.7774
4	4	0.06857	0.802
5	5	0.06857	0.2302
6	6	0.06857	0.528
7	7	0.06857	0.8124
8	8	0.06857	0.7812
9	9	0.06857	0.7507
10	10	0.06857	0.0974
11	12	0.06857	0.8617
12	16	0.06857	0.8506
13	21	0.06857	0.1932
14	27	0.06857	0.8378
15	34	0.06857	0.5048
16	44	0.06857	0.5814

Least common period count: 1

Add
Remove
Clear
Import...
Export...
Preview...
Signal optimization...
Reset all phases to zero

→ Excitation signal wizard...
The excitation wizard will help you to generate an excitation signal according to your demands

OK Cancel

Figure 30: Excitation Signal Editor Dialog.

In the Excitation Signal Editor, you can adjust the frequency table for your signal excitation. To get started, a predefined frequency table is stored on the device and can also be found in the *Frequency Table* folder on the delivered USB-drive.

Within the Excitation Signal Editor, you can add or remove single frequency points of the multi-sine signal from the table, export the frequency table, or import one. Each frequency point can then be defined by its frequency in Hertz, Amplitude in Ampere, and a phase shift in **per-unit value** ($0 \leq \text{phase} \leq 1$).

Item	Description
Preview...	Open the preview window to show a visualization of the signal.
Signal optimization...	Open the Signal Optimization window to enable several optimization options.
Reset all phases to zero	Set the phase angles to 0 for all frequencies.
Excitation signal wizard	Start a wizard that helps you to generate a suitable current excitation signal.

Finish the editing of the frequency table by pressing the *OK* button. If the start point has not been optimized and the initial current value of the signal is greater than 1 % of the total signal amplitude, a notification suggests an automatic phase correction as described above.

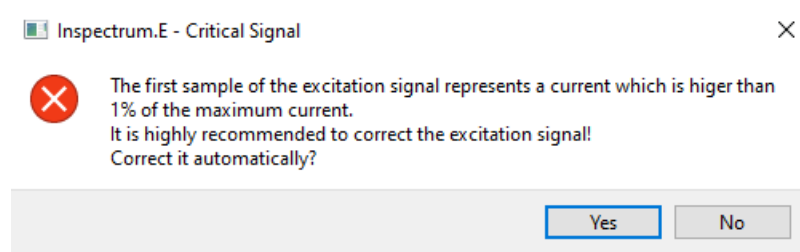


Figure 31: Warning popup for initial signal amplitude.

10.1.2 Signal optimization

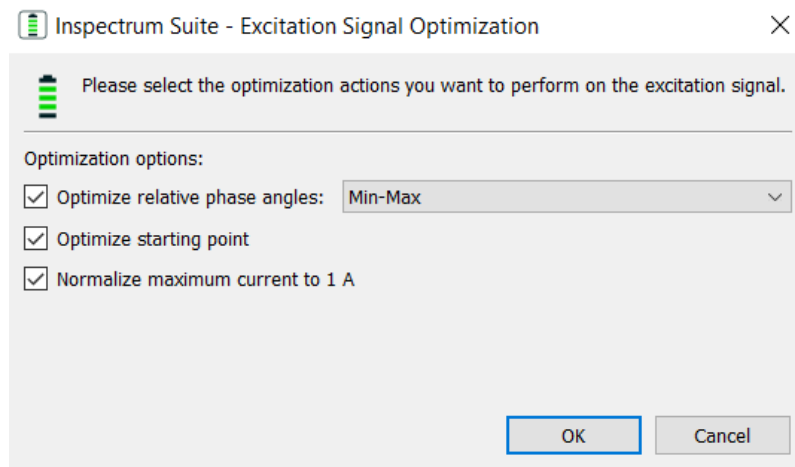
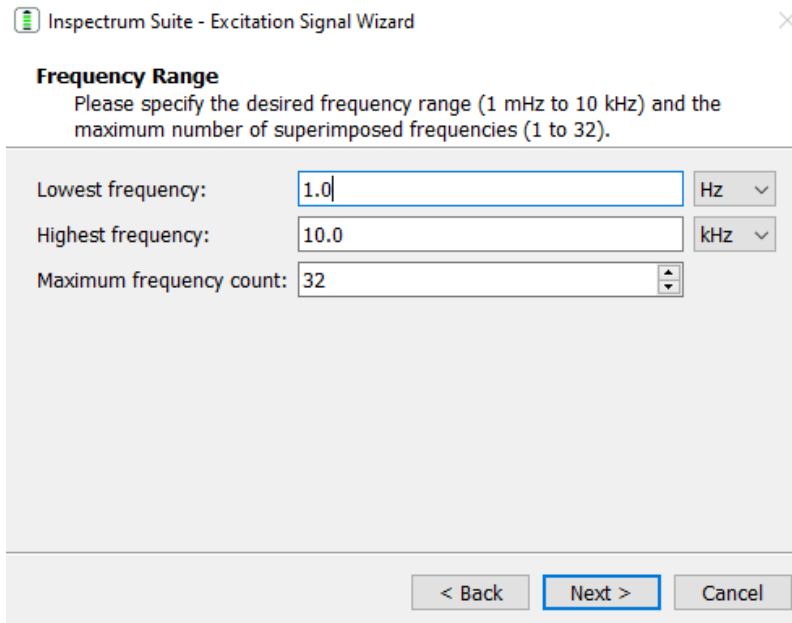


Figure 32: Excitation Signal Optimization Dialog.

Item	Description
Optimize relative phase angles:	This function optimizes the phase angles of all frequency components to spread the signal more evenly over the complete excitation duration. Select "Optimize Crest / Random Phase" to optimize for a low crest factor from a random selection of phase angles. Alternatively, the "Min-Max" option can be used, which typically provides a similarly accurate result with shorter calculation time. The "Fast Min-Max" option will yield an even quicker result, but it may not be optimal in every case. The "Optimize for single-period" option is designed specifically for measurements with only a single excitation period. To configure the maximum time spent optimizing the signal, click on the <i>Preferences</i> button in the toolbar and move to the <i>Misc</i> tab.
Optimize starting point:	Enable this function to automatically shift the signal phase to an initial current zero-crossing. This shift is necessary if the signal's initial current value is greater than 1 % of the peak current value. Enabling this function avoids initial current discontinuities without altering the signal. Select the "Fast" option for a quick result, or the "Precise" option for a more accurate optimization.
Normalize to 1 A:	Enable this function to normalize the total signal amplitude to 1 A. By doing so, the signal's desired excitation amplitude in Ampere can be directly set by the parameter <i>Current scaling factor</i> field in the <i>Excitation</i> tab.

10.1.3 Excitation Signal Wizard (Pro version only)

Follow the instructions of the Excitation Signal Wizard to manually set the frequency range, number of frequency counts, optimization strategy, and other relevant parameters.



Frequency Range
Please specify the desired frequency range (1 mHz to 10 kHz) and the maximum number of superimposed frequencies (1 to 32).

Lowest frequency: Hz

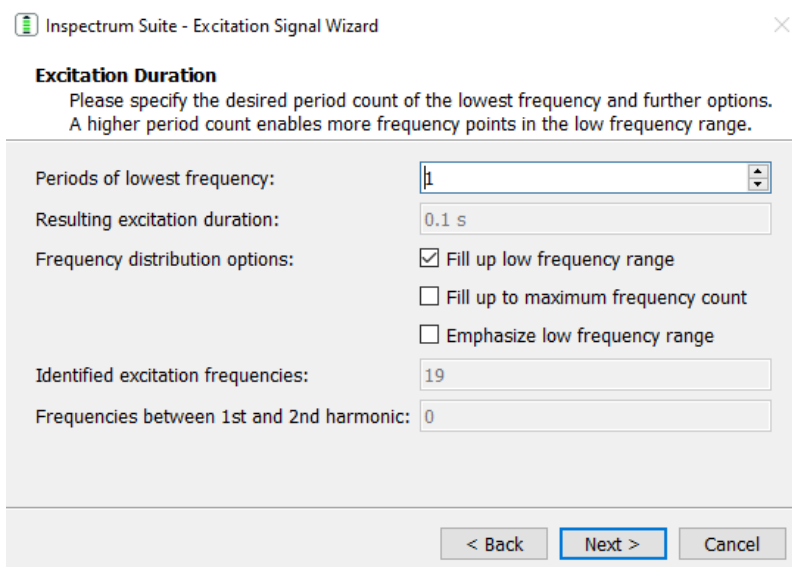
Highest frequency: kHz

Maximum frequency count:

< Back **Next >** Cancel

Figure 33: Excitation Signal Wizard: Step 1.

Frequency Range: This allows you to specify the desired frequency range, between 1 mHz and 10 kHz, and the maximum number of superimposed frequencies, up to 32 frequencies.



Excitation Duration
Please specify the desired period count of the lowest frequency and further options.
A higher period count enables more frequency points in the low frequency range.

Periods of lowest frequency:

Resulting excitation duration:

Frequency distribution options:

- ☒ Fill up low frequency range
- ☐ Fill up to maximum frequency count
- ☐ Emphasize low frequency range

Identified excitation frequencies:

Frequencies between 1st and 2nd harmonic:

< Back **Next >** Cancel

Figure 34: Excitation Signal Wizard: Step 2.

Excitation Duration: This allows you to specify the period count of the lowest frequency. A higher period count enables more frequency points in the low-frequency range. The resulting excitation duration for the specified value is shown directly below.

With the frequency distribution options, you can tune the allocation of available frequencies until the maximum is reached. If the option *Fill up to maximum frequency count* is selected, this maximum is 32 frequencies. Otherwise, the frequency count selected in the previous step will be used. For an especially high emphasis on low frequencies, the option *Emphasize low frequency range* should be selected.

Please note that these options change the resulting frequency table if the maximum is not already exhausted by the previous step.

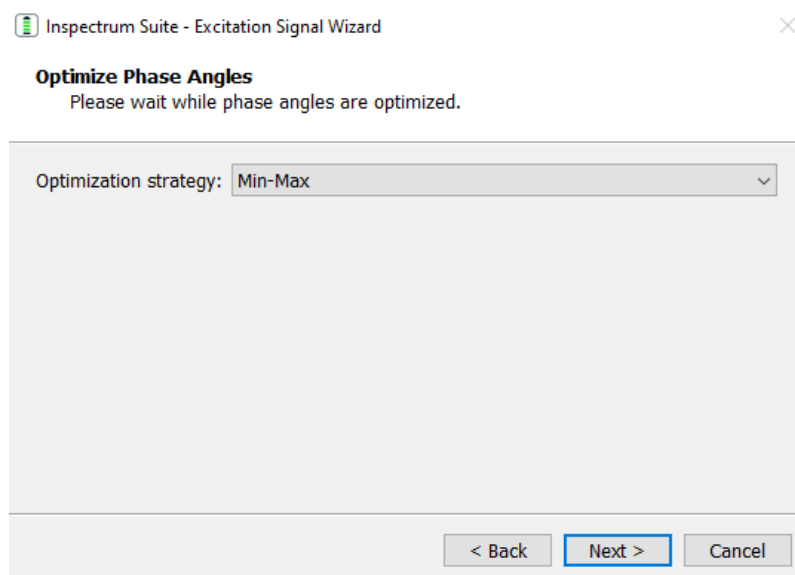


Figure 35: Excitation Signal Wizard: Step 3.

Optimize Phase Angles: This function optimizes the phase angles of all frequency components to spread the signal more evenly over the complete excitation duration. Select *Optimize Crest / Random Phase* to optimize for a low crest factor from a random selection of phase angles. Alternatively, the *Min-Max* option can be used, which typically provides a similarly accurate result with shorter calculation time. The "Fast Min-Max" option will yield an even quicker result, but it may not be optimal in every case. The "Optimize for single-period" option is designed specifically for measurements with only a single excitation period. To configure the maximum time spent optimizing the signal, click on the *Preferences* button in the toolbar and move to the *Misc* tab. The *No optimization / All phases zero* option will skip this step and set all phase angles to 0.

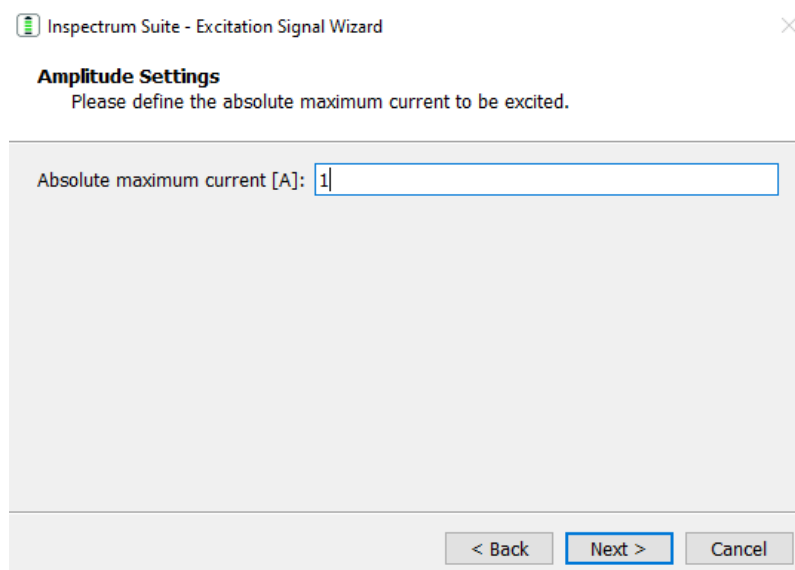


Figure 36: Excitation Signal Wizard: Step 4.

Amplitude Settings: This allows you to specify the absolute maximum current of excitation.

Important information

Caused by the nature of the multi sine measurement mode, the number of periods every single frequency is excited depends mainly on the lowest frequency.

As the Inspectrum covers a great range of possible frequencies, the user must pay attention that the maximum period count of 100,000 is not exceeded for any single frequency.

Using a frequency table that contains frequencies from 0.1 Hz to 10 kHz will result in a period count of 100,000 for the highest frequency, which is just within the technical limit.

Surpassing this limit may cause inaccurate measurement results.

Consequently, the full frequency range of the Inspectrum should not be used in one multi sine measurement. If it is necessary to use the full frequency range, the **multi sine sweep functionality** can be used.

10.2 Limits

Device Configuration

×

ExcitationLimitsMultiplexerCAN InterfaceMisc.Device Information

Over-current limit [A]:2.5

Over-voltage limit [V]:4.2

Under-voltage limit [V]:0

Tolerable fault duration [ms]:10

→ Store parameters in EEPROM

Parameter values will be reloaded after device reset or power cycle

Figure 37: Device configuration: Limits.

Parameter	Range
Over-current limit [A]	0 - 5 A (Amplitude)
Over-voltage limit [V]	-0.5 - 5.2 V
Under-voltage limit [V]	-0.5 - 5.2 V
Tolerable fault duration [ms]	Set the time interval for which one of the error cases mentioned above needs to occur before a measurement is aborted

You can find this information in the datasheet of your cell.

10.3 Multiplexer

Device Configuration ✕

ExcitationLimitsMultiplexerCAN InterfaceMisc.Device Information

Channel sequencer:☐ Channel sequencer enabled

Scaling factor sequencer:☐ Scaling factor sequencer enabled

Channel sequence:

1;2;3;

Scaling factor sequence:

Edit multiplexer sequence...

→ Store parameters in EEPROM

Parameter values will be reloaded after device reset or power cycle

Figure 38: Device configuration: Multiplexer.

Item	Description
Channel sequencer	Enable this function to allow the Inspectrum.C to cycle through the specified multiplexer channels during sweep - and continuous measurement mode.
Scaling factor sequencer	Enable this function to allow the sequencing of the scaling factor if the channel sequencer is enabled. If disabled, the Scaling Factor specified in the Excitation tab will be used.

Item	Description
Channel sequence	Shows your resulting channel sequence based on your settings for the multiplexer sequence.
Scaling factor sequence	This shows your resulting scaling factor sequence based on your settings for the multiplexer sequence.
Edit multiplexer sequence...	Open the Multiplexer Sequence window to configure the measurement sequence of the multiplexer.

10.3.1 Multiplexer Sequence

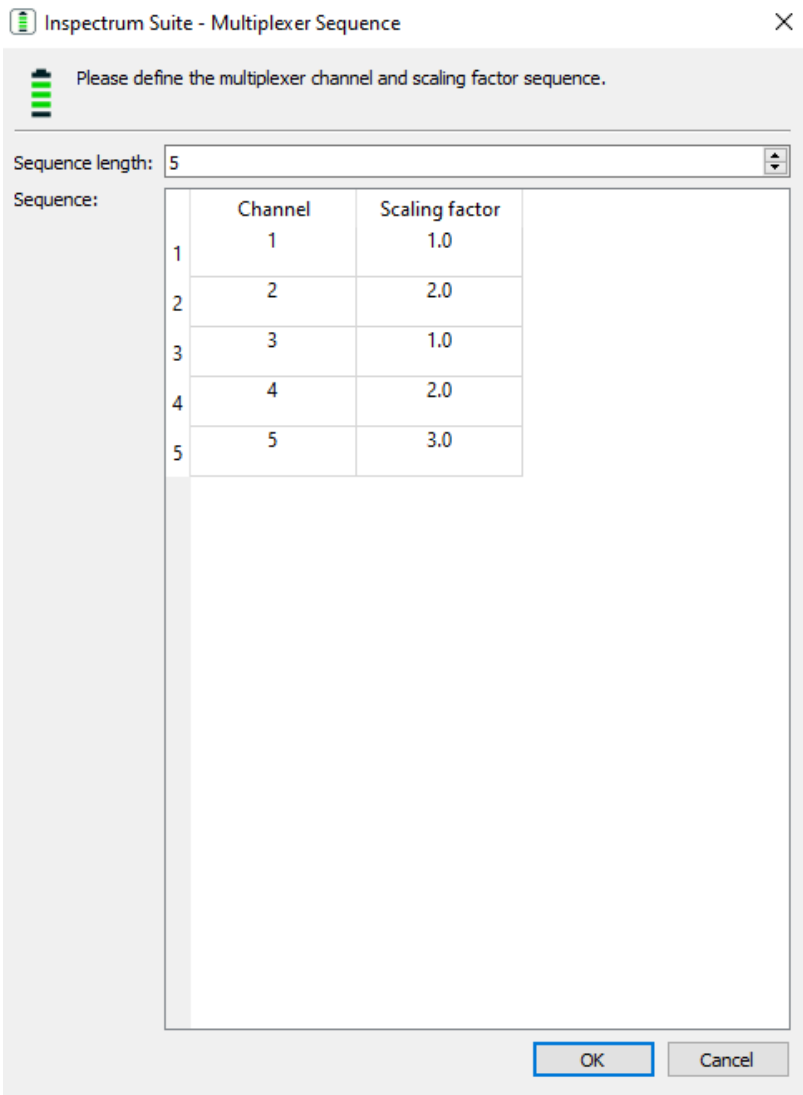


Figure 39: Device configuration: Multiplexer Sequence Editor.

In the Multiplexer Sequence window, you can specify the sequence length of your multiplexer measurement and adjust the channel number and scaling factor for each respective channel, if necessary. After pressing the *OK* button, the configuration will be shown in the *Channel sequence* and *Scaling factor sequence* fields of the *Multiplexer* tab.

Multiplexer Sequence Length

The maximum allowed sequence length depends on the used device type and hardware revision.

10.4 CAN Interface

For instructions on how to work with the CAN Interface, please see the [corresponding chapter](#).

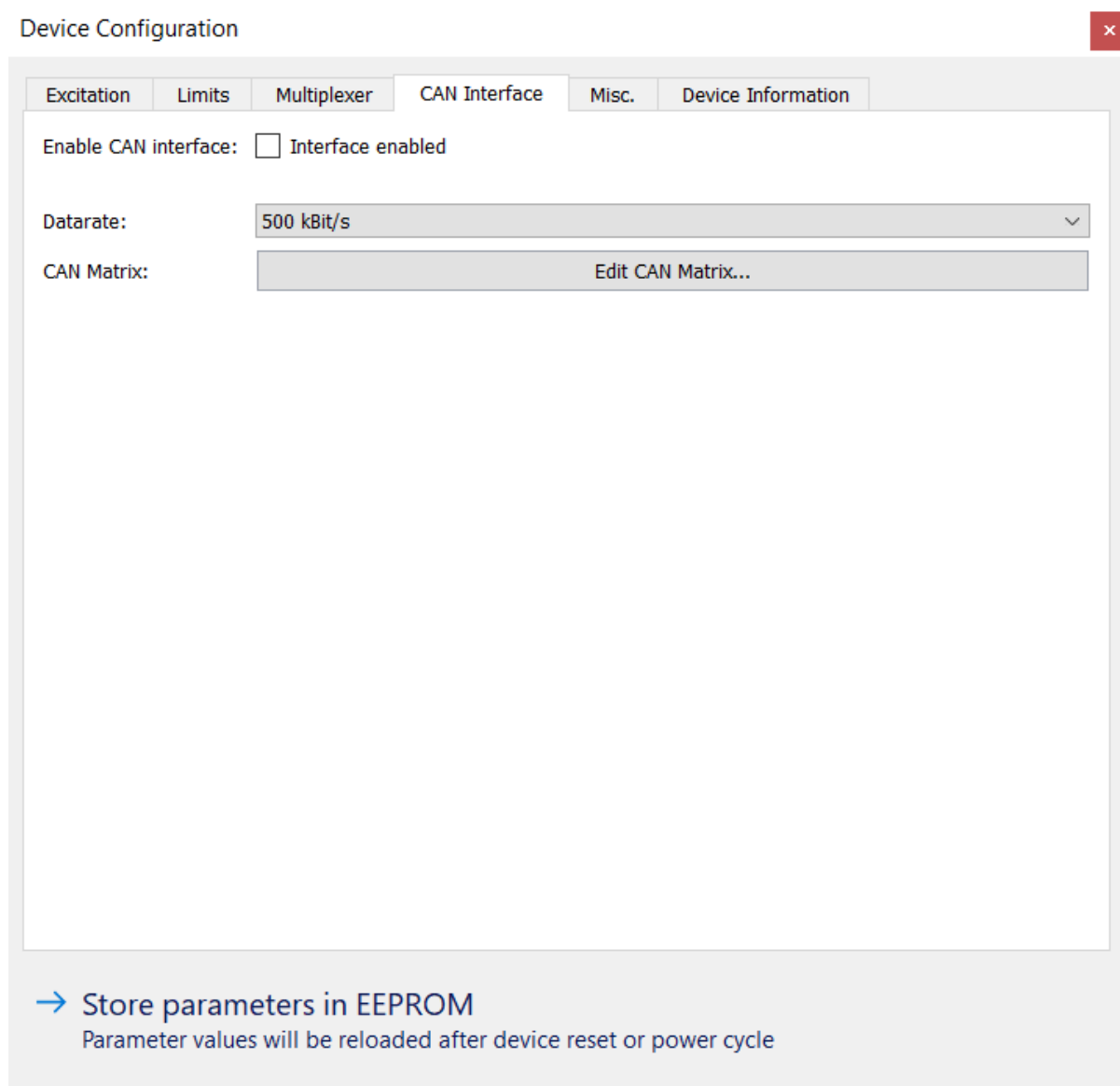


Figure 40: Device configuration: CAN interface.

Item	Description
Enable CAN interface	Enable this function to activate the CAN interface on the device (indicated by the reaction of the CAN status LEDs on transmitted or received messages).
Datarate	Choose the maximum data rate for the CAN interface.
CAN Matrix	Click on "Edit CAN Matrix..." to open the CAN Matrix Editor for the Configuration of the CAN communication.

10.4.1 CAN Matrix Editor

Inspectrum Suite - CAN Matrix Editor

Please edit the CAN identifier of each CAN message according to your demands. Check the message to enable the reception or transmission on the Inspectrum device, uncheck to disable.

Messages to Inspectrum:		Messages from Inspectrum:	
Message:	Identifier:	Message:	Identifier:
☒ Connection Status Read Write	0x101	☒ Connection Status Value	0x102
☒ Device Status Read Write	0x103	☒ Device Status Value	0x104
☒ Parameter Read Write	0x105	☒ Parameter Value	0x106
☒ Error Read Reset	0x107	☒ Error Value	0x108
☒ Measurement Result Read	0x109	☒ Measurement Result Value	0x10a
☒ Multiplexer Status Read Write	0x10b	☒ Multiplexer Status Value	0x1c

User mailboxes (to Inspectrum):	
☒ User Mailbox 1	0x10d
☒ User Mailbox 2	0x10e
☒ User Mailbox 3	0x10f
☒ User Mailbox 4	0x110

→ Setup in ascending order
IDs will be generated based on the "Connection Status Read Write" identifier

OK Cancel

Figure 41: Device configuration: CAN matrix editor.

Within the CAN Matrix Editor, you can enable/disable control messages and set your CAN identifiers. The interface communicates with a standard 11-Bit identifier.

For convenience, if a specific identifier range is available for the Inspectrum.C, you can simply insert the lowest available identifier into the *Connection Status Read Write* input field and click the *Setup in ascending order* button. Based on the first element, this will set all other identifiers in an increasing order.

Please note that the USB and CAN interface are mutually exclusive. Thus, the USB Connection must be closed before establishing the connection via CAN. An exception is the four user mailboxes available in the CAN Matrix Editor. With the help of these, the Inspectrum.C software can work as a data logger for additional meta data provided over the CAN-Bus.

These mailboxes are user-specific messages with an 8 Byte long data field. after each measurement, all data transmitted to the mailboxes will be written into the XML result files. These mailboxes allow for tracking of external data like the current and temperature of an external battery cyler within one file. The mailboxes are always active when the CAN interface is enabled, independent of the CAN connection status.

A full description of the CAN interface can be found in the [corresponding chapter](#).

10.5 Miscellaneous

Device Configuration

Excitation

Limits

Multiplexer

CAN Interface

Misc.

Device Information

Automatically connect to battery:

☒ Enable automatic connect

Automatically disconnect from battery:

☒ Enable automatic disconnect

Enter power saving mode after [s]:

5000

→ Store parameters in EEPROM

Parameter values will be reloaded after device reset or power cycle

Figure 42: Device configuration: Misc. settings

Item	Description
Automatically connect to battery	Enable this function to automatically connect a cell as soon as a measurement is initiated.
Automatically disconnect from the battery	Enable this function to disconnect a cell directly after a measurement automatically. This way, the discharge of the battery cell due to small leakage currents is minimized in case of long measurement intervals.
Enter power saving mode after [s]	Set the time for an automatic power-down duration by entering a non-zero time period. If no measurement is taking place during this time interval, the Inspectrum.C disconnects the battery and goes into a low power state.

10.6 Device Information

Device Configuration

ExcitationLimitsMultiplexerCAN InterfaceMisc. Device Information

Hardware revision:1.0

Software revision:1.4

Serial number:IE10522020011

Calibration Date:2022/03/02

→ Store parameters in EEPROM

Parameter values will be reloaded after device reset or power cycle

Figure 43: Device configuration: Device Information

Figure 44:

View your device\'s hardware and software version, serial number, and the calibration date. Depending on the connected device and the software version, additional device hardware features can also be unlocked here.

11 Preferences

To change the software preferences, click on the *Preferences* button in the toolbar.



11.1 Workspace

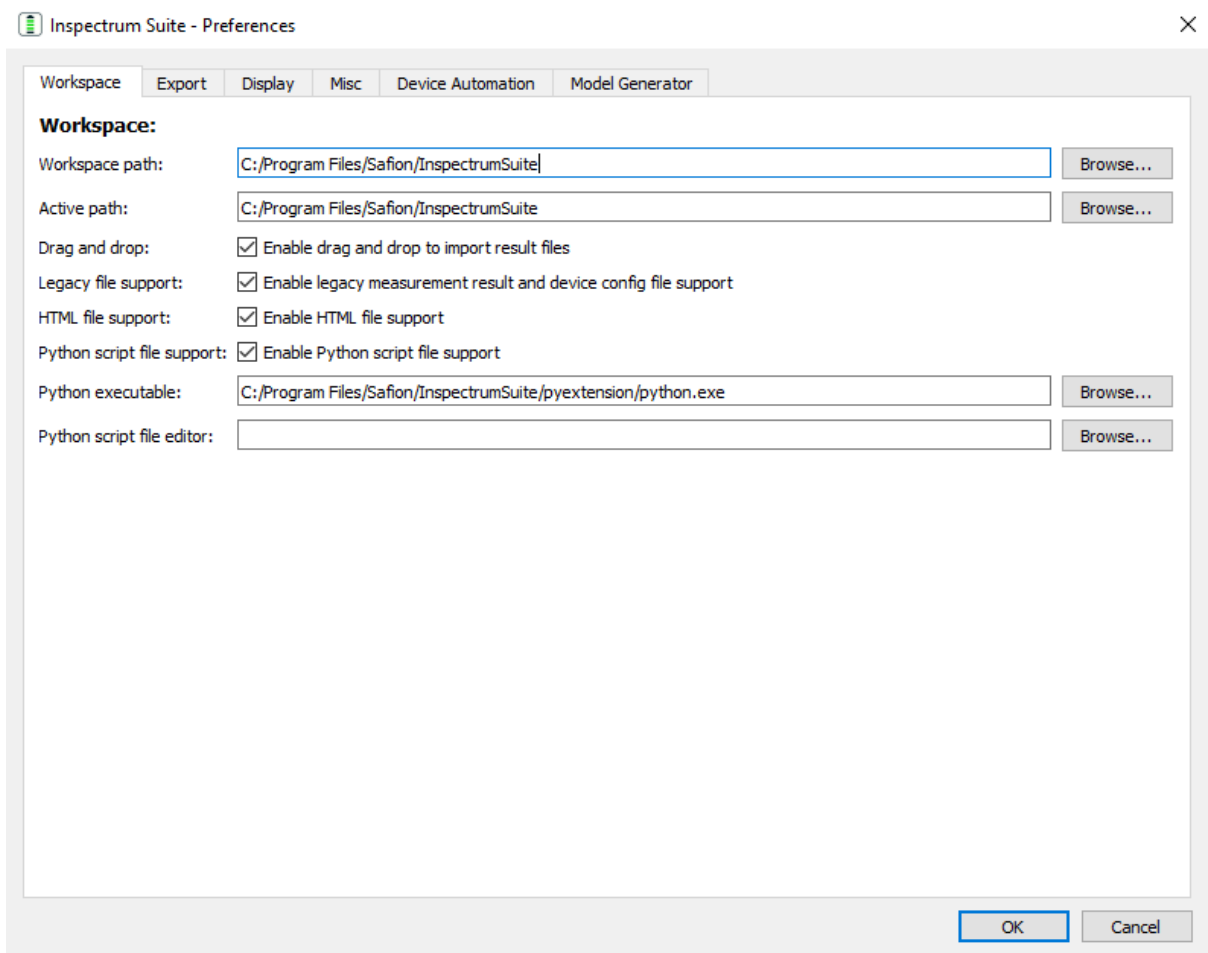


Figure 45: Preferences: Workspace.

Item	Description
Workspace path	Set the path for your workspace. This path is the root of the tree view in the Workspace Explorer.

Item	Description
Active path	Set the active path for your current measurement. Here all your files will be exported. This path needs to be either your workspace path, or a subfolder within your workspace.
Drag and drop	Enable this function to allow importing measurement results files from the Workspace Explorer via drag-and-drop.
Legacy file support	Enable this function to support older file formats.
HTML file support	If this function is enabled, .html files will be displayed in the workspace explorer and can be opened in the default browser by double-clicking on them.
Python script file support	If this function is enabled, .py files will be displayed in the workspace explorer.
Python executable	Specify the path where the Python executable is located, to be able to run Python files.
Python script file editor	Specify the path where the Python script file editor is located. Double-clicking the .py file in your Workspace Explorer will automatically open it in the selected editor.

11.2 Export


Inspectrum Suite - Preferences
×

Workspace
Export
Display
Misc

Export:

Export file selection dialog: ☒ Show file selection dialog on non-automatic result export

Automatic export: ☒ Enable automatic result export

Burst measurement mode: ☒ Export burst measurement results in one single file

Result files will be exported to active path with the following properties:

File name:

Multiplexer channel: ☒ Include multiplexer channel in file name

Suffix counter digits:

Resulting file name:

Comment:

Battery names:

Include sections:

- ☒ Multiplexer configuration
- ☐ CAN configuration
- ☐ Misc. configuration

Figure 46: Preferences: Export.

Item	Description
Export file selection dialog	Enable this function to show a dialog window for manually choosing between files when exporting.
Automatic export	Enable this function to export your measurement results after the measurement is completed automatically.
Burst measurement mode	Enable this function to export your burst measurement results in one file.
File name	Set a custom file name prefix for exported measurement results.
Multiplexer channel	Enable this function to always include the multiplexer channel in the file name.

Item	Description
Suffix counter digits	Define the number of digits for the suffix counter in the file name.
Resulting file name	View the resulting file name based on the settings above.
Comment	Include a comment in exported files.
Battery names	Click on "Setup battery names" to open the Battery Names Setup window for defining battery names for the measurement results export.
Include sections	Chose the categories for device configuration exports.

11.2.1 Battery Names Setup

In the Battery Names Setup dialog, you can set a specific name for the connected battery that will be saved in the measurement data. These names are especially useful in a multiplexed measurement setup to differentiate each battery in your exported data.

Inspectrum Suite - Battery Names Setup

Please specify a battery name or a set of battery names for a multiplexed measurement setup.

Battery name for a non-multiplexed measurement setup (Channel 0):

No Multiplexer (channel 0): Test0

Battery names for a multiplexed measurement setup:

Multiplexer channel 1: Test1

Multiplexer channel 2:

Multiplexer channel 3:

Multiplexer channel 4:

Multiplexer channel 5:

Multiplexer channel 6:

Multiplexer channel 7:

Multiplexer channel 8:

OK Cancel

Figure 47: Preferences: Battery Names.

11.3 Display

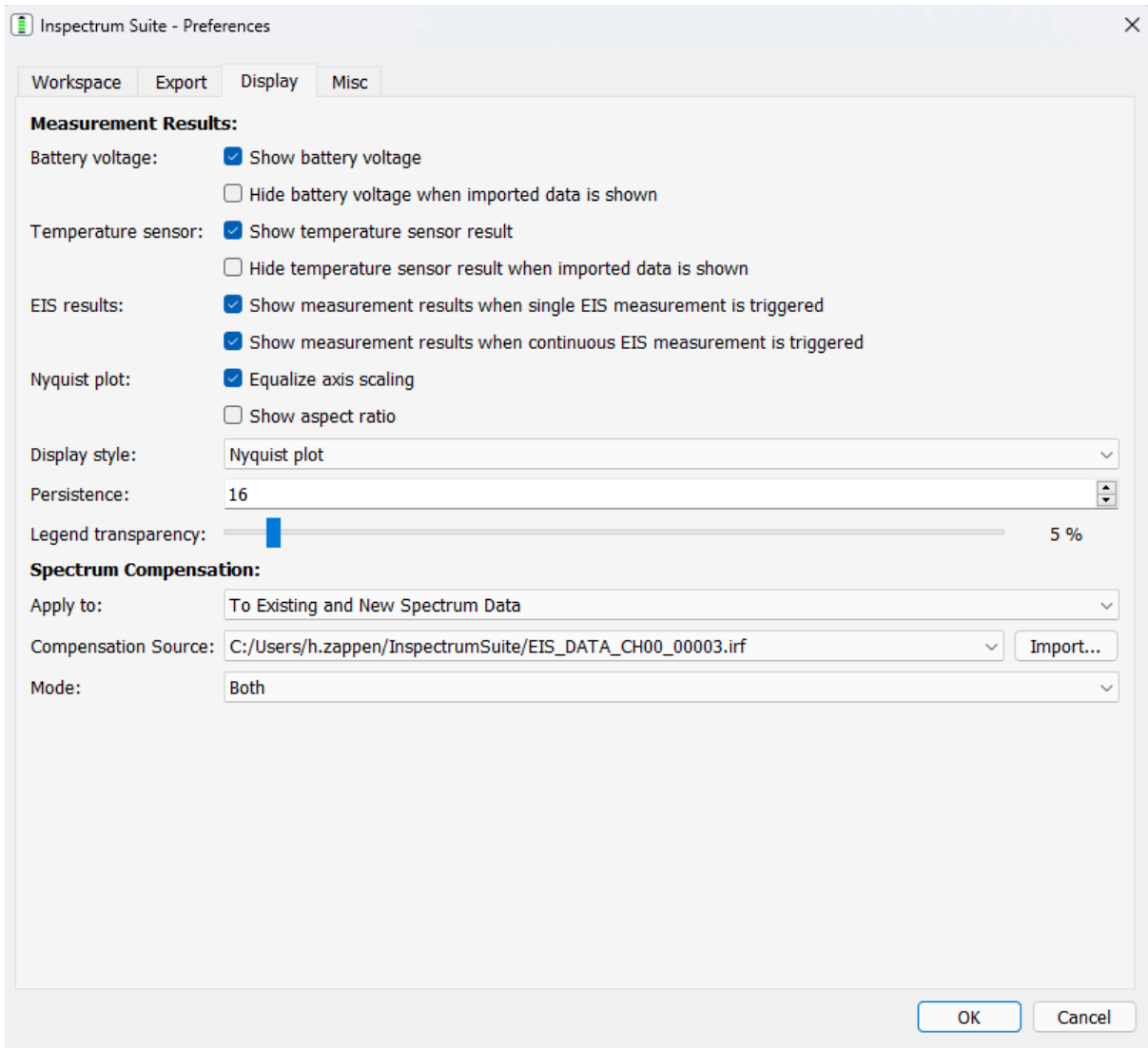


Figure 48: Preferences: Display.

Item	Description
Battery voltage	Show/hide the battery voltage live data in the Main Area.
Temperature sensor	Show/hide live temperature data in the Main Area.
EIS results	Set if measurement results should be shown after a measurement is triggered for single and continuous EIS measurements.
Nyquist plot	Enable the equalization of axis scaling, and show the aspect ratio for Nyquist plots.

Item	Description
Display style	Chose between Nyquist plot and Bode plot for the measurement results visualization.
Persistence	Set the number of traces displayed in the measurement results visualization.
Legend transparency	Adjust the transparency of the legend in the measurement results visualization.

Additionally, the spectrum compensation function can also be configured from here. Please refer to the [corresponding chapter](#) for an overview of the options.

11.4 Miscellaneous

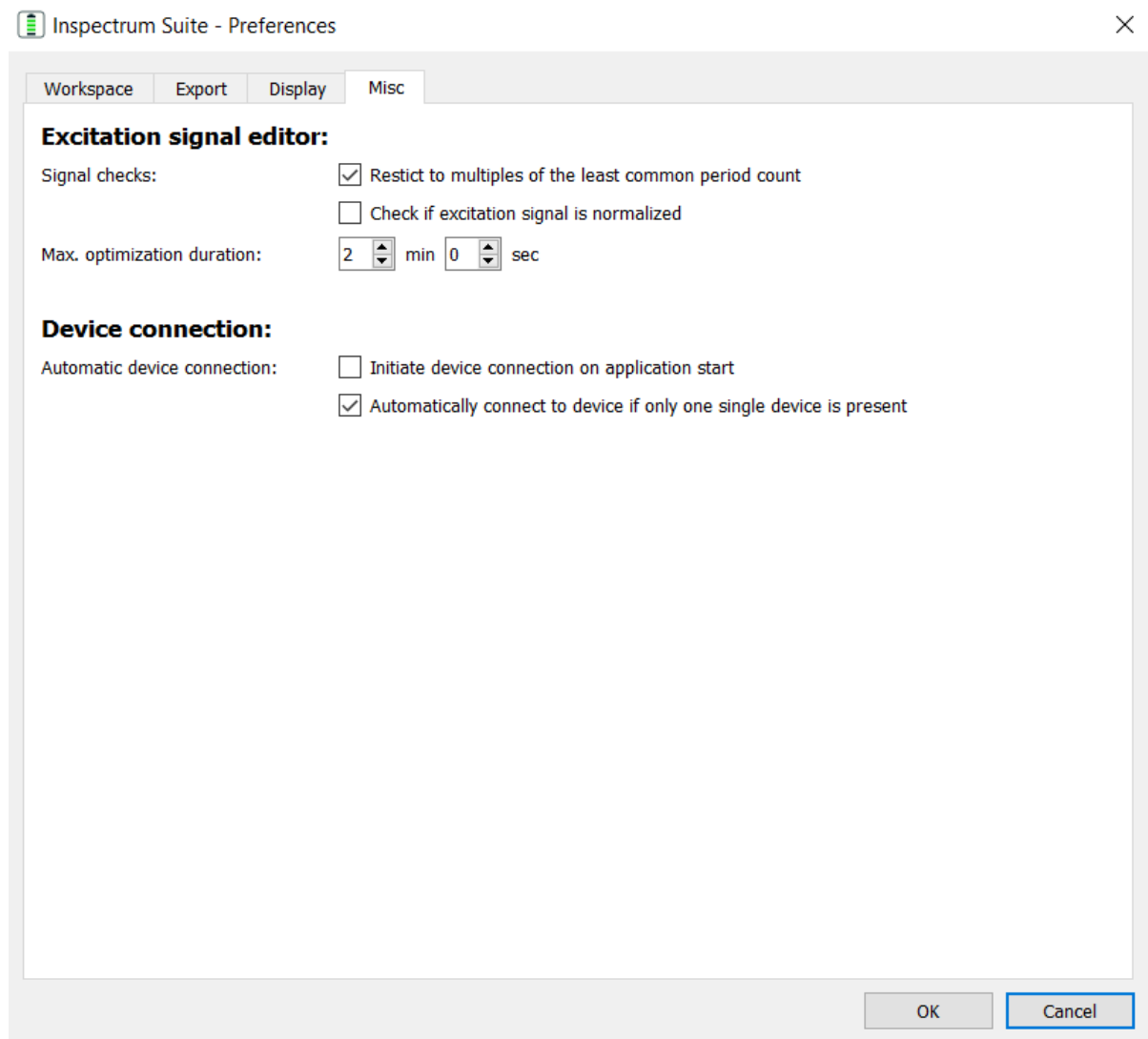


Figure 49: Preferences: Miscellaneous.

Parameter	Description
Excitation signal editor	Enable/disable the restriction to multiples of the least common period count. Enable/disable automatic checking if the excitation signal is normalized. Specify the maximum optimization duration.
Device connection	Enable/Disable automatic opening of device connection wizard when starting the software application. You can also Enable/Disable automatic connection, if only a single device is plugged in.

12 CAN Interface

Besides the USB Interface, the Inspectrum.C can also be controlled via a CAN-Interface with a slightly reduced instruction set. This instruction set includes the starting and stopping of measurements, the setup of specific parameters, and the readout of measurement results.

To enable the CAN interface, click on the *Device configuration* button in the toolbar and move to the *CAN Interface* tab. For a detailed explanation of these configuration options, please refer to the [device configuration chapter](#).

 Important information

- Automatic multiplexer channel sequencing is not supported by the CAN interface and must be done via USB
- In the external CAN device's system, a protocol must be implemented to allow the control of Inspectrum.C. This protocol will be explained in the following section.

CAN Protocol

As a reference, the structure of the CAN protocol is provided via an Excel table and a DBC file in the software folder. For simplicity, only the excel table will be referred to in this manual. The different layers of the protocol, which will be described later, are arranged into different excel sheets at the bottom left corner:

Command Set	Multiplexer Status IDs	Device Status IDs	Result IDs	Parameter IDs	Error Codes
--------------------	------------------------	-------------------	------------	---------------	-------------

To begin with, the CAN protocol is built upon a set of Read-Write-Request messages and their corresponding reply messages. This set of messages represents the device\'s control groups, including the connection control, device control, parameter setup, error status, and multiplexer control. In addition, there are dedicated CAN-IDs for the measurement result values and user mailboxes. Within these control group messages, different functions are multiplexed with an individual parameter ID value in the data field of the CAN message.

Generally, with the Read-Write-Request messages, the direction of the data flow can be controlled. As shown in the table below, in the case of a write action (Data 0 = 1), the desired parameter ID and parameter value has to be transmitted in the remaining data fields of the message. Vice versa, only the desired parameter ID must be specified if the user requests a read action (Data 0 = 0). The Inspectrum.C will then respond with the requested parameter ID and value. Whenever there is an internal status change in the Inspectrum.C (for example, after a parameter write or an error occurred), the Inspectrum.C will send a status message with the updated parameter.

	Data 0	Data 1	Data 2 - 5
Write Request	1 (Write)	Parameter ID	Parameter Value
Read Request	0 (Read)	Parameter ID	-

	Data 0	Data 1	Data 2 - 5
Read Reply / Status Update	Parameter ID	Parameter Value	Parameter Value

To take the interface into operation, the CAN connection must be opened with the corresponding **CONNECTION_STATUS_READ_WRITE** command, requesting a write and a connection opening process:

Command	Data 0	Data 1
Write Request	CONNECTION_STATUS_READ_WRITE	1 (Write)

Next, the device will respond with a **CONNECTION_STATUS_VALUE** message confirming the successful connection:

Command	Data 0	Data 1
Status Reply	CONNECTION_STATUS_VALUE	1 (Open)

If the USB interface gets connected or the Power Button gets pressed while the CAN Interface is enabled, the CAN interface gets disconnected. The status change gets transmitted via the **CONNECTION_STATUS_VALUE** message.

Command	Data 0	Data 1
Status Reply	CONNECTION_STATUS_VALUE	0 (Closed)

The device can now control the Inspectrum.C via the CAN interface. With the **PARAMETER_READ_WRITE** command, reduced configuration settings can be transmitted or read out. For this, the user must select the desired parameter in the *Parameter IDs* tab of the excel table. The ID, unit, byte count and datatype are specified in the table for each available parameter. The parameter ID and value can then be inserted into the data field of the **PARAMETER_READ_WRITE** command requesting a write action. An example of the configuration of the scaling factor is given below. All parameter values are transmitted in big-endian notation.

	Command	Data 0	Data 1	Data 2-6
Write Request	PARAMETER_READ_WRITE	1 (Write)	0x11 (Scaling Factor)	[0x3F][0x19][0x99][0x9A] (Value = 0.6)

The Inspectrum.C will respond with a **PARAMETER_VALUE** message to confirm the parameter modification:

	Command	Data 0	Data 1-5
Status Reply	PARAMETER_VALUE	0x11 (Scaling Factor)	[0x3F][0x19][0x99][0x9A] (Value = 0.6)

By sending the **PARAMETER_READ_WRITE** command with a read request (Data 0 = 0), the status of a certain parameter can also be requested. Specific device parameters such as the multiplexer channel and scaling factor sequences are not supported via the CAN Bus. These settings must be executed via the USB interface before establishing the CAN connection.

After the configuration is finished, the battery can be connected, and the measurement can be triggered. All device control functions are grouped within the *Device Status IDs* tab of the excel sheet. Like the parameter modification, the **DEVICE_STATUS_READ_WRITE** command can be used to transmit or read out the device control flags. The same holds true for the multiplexer configuration and channel select, grouped into the *Multiplexer Status IDs* tab. An Example of a measurement starting process is given below:

	Command	Data 0	Data 1	Data 2
Write Request	MULTIPLEXER_STATUS_READ_WRITE	1 (Write)	0x10 (Channel Select)	[0x01] (Channel 1)
Write Request	DEVICE_STATUS_READ_WRITE	1 (Write)	0x20 (Connect Battery)	[0x01] (Connect)
Write Request	DEVICE_STATUS_READ_WRITE	1 (Write)	0x02 (EIS Measurement)	[0x0] (Start)

As a response, the Inspectrum.C will reply with the corresponding updated Status Flags, indicating that the measurement has been started.

	Command	Data 0	Data 1
Status Reply	DEVICE_STATUS_VALUE	0x10 (Channel Select)	[0x01] (Channel 1 selected)
Status Reply	DEVICE_STATUS_VALUE	0x20 (Connect Battery)	[0x01] (Connected)
Status Reply	DEVICE_STATUS_VALUE	0x02 (EIS Measurement)	[0x1] (Measurement Running)

As soon as the EIS measurement is complete, the Inspectrum.C will signal the status change with a device status message:

	Command	Data 0	Data 1
Status Reply	DEVICE_STATUS_VALUE	0x02 (EIS Measurement)	[0x0] (Measurement Finished)

The user can now read the measurement result values with the **MEASUREMENT_RESULT_READ** command set. Each measurement result value is linked to a result ID as provided in the excel sheet. An example of the readout of the Clipping Flag is given below:

	Command	Data 0	Data 1
Read Request	MEASUREMENT_RESULT_READ	0 (Read)	0x14 (Clipping Flag)

The Inspectrum.C replies with a **MEASUREMENT_RESULT_VALUE**. This message contains an additional field for the multiplexer channel which has been active during the measurement:

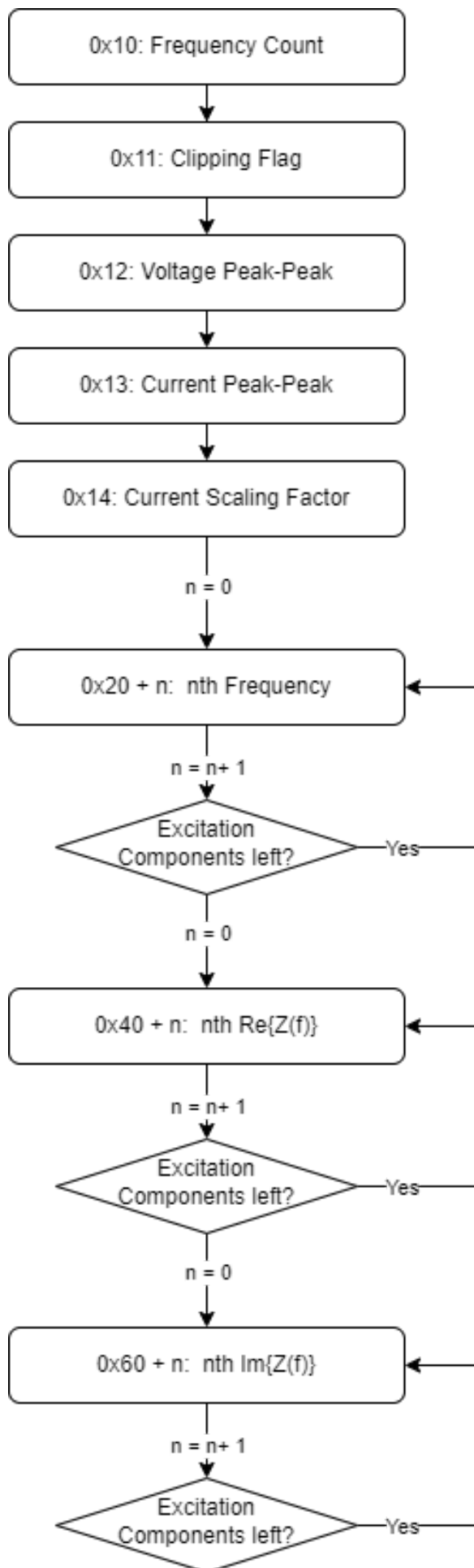
	Command	Data 0	Data 1	Data 2	Data 3-5
Status Reply	MEASUREMENT_RESULT_VALUE	0x14 (Result ID)	1 (Mux. Channel)	0 (No Clipping)	-



If the *Send results when measurement finished* option in the device parameters (Parameter ID = 0x80) has been activated, the Inspectrum.C is also able to automatically transmit all measurement results after a complete measurement without the need for user intervention.

The sequence of the result IDs is shown in the following figure. After a couple of excitation meta-data messages, the device will loop through all frequency components of the excitation spectrum, transmitting the frequency, real part, and imaginary part of the battery impedance. The sequence is finished with an *End of Spectrum* message.

In case of the battery voltage and temperature measurement are enabled, the results of these measurements will also be transmitted with their corresponding result ID.



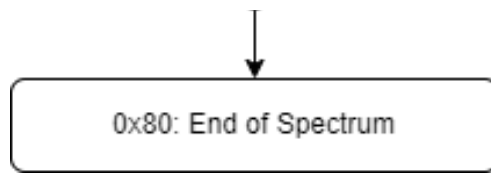


Figure 50: CAN Bus interface: Command flow.

I.I Miscellaneous

13 Firmware Update

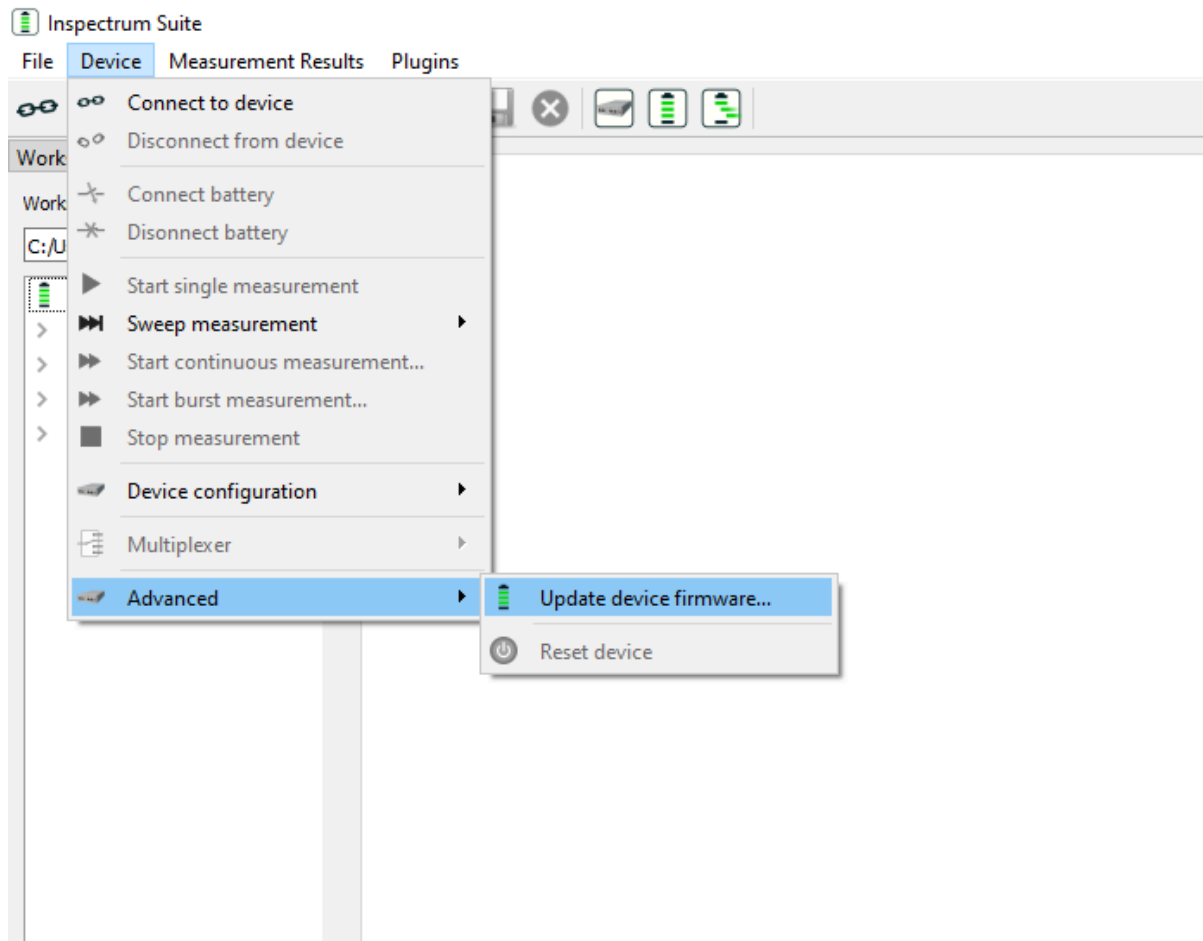


Figure 51: Firmware Update via Menu.

The Inspectrum.C firmware can be updated manually with a corresponding .bin firmware file. Make sure the Inspectrum.C is connected to your computer via USB before trying to install the firmware update. Click on *Device* → *Advanced* → *Update device firmware...*, then browse your local drives for the firmware file and click on *Start firmware update...* to begin installing.

The Active LED on the front panel of the Inspectrum.C will start blinking to signal that the update is in progress. The process may take a few minutes and, once completed, the new firmware will be started automatically.

Warning

Do not turn off the Inspectrum.C or your computer during installation to avoid corrupting the firmware and having to restart the process.

14 Plugins (Pro version only)

In addition, further functionality is provided with user-selectable plugins. The plugin manager is accessible by clicking the "Plugin Manager" button in the toolbar.

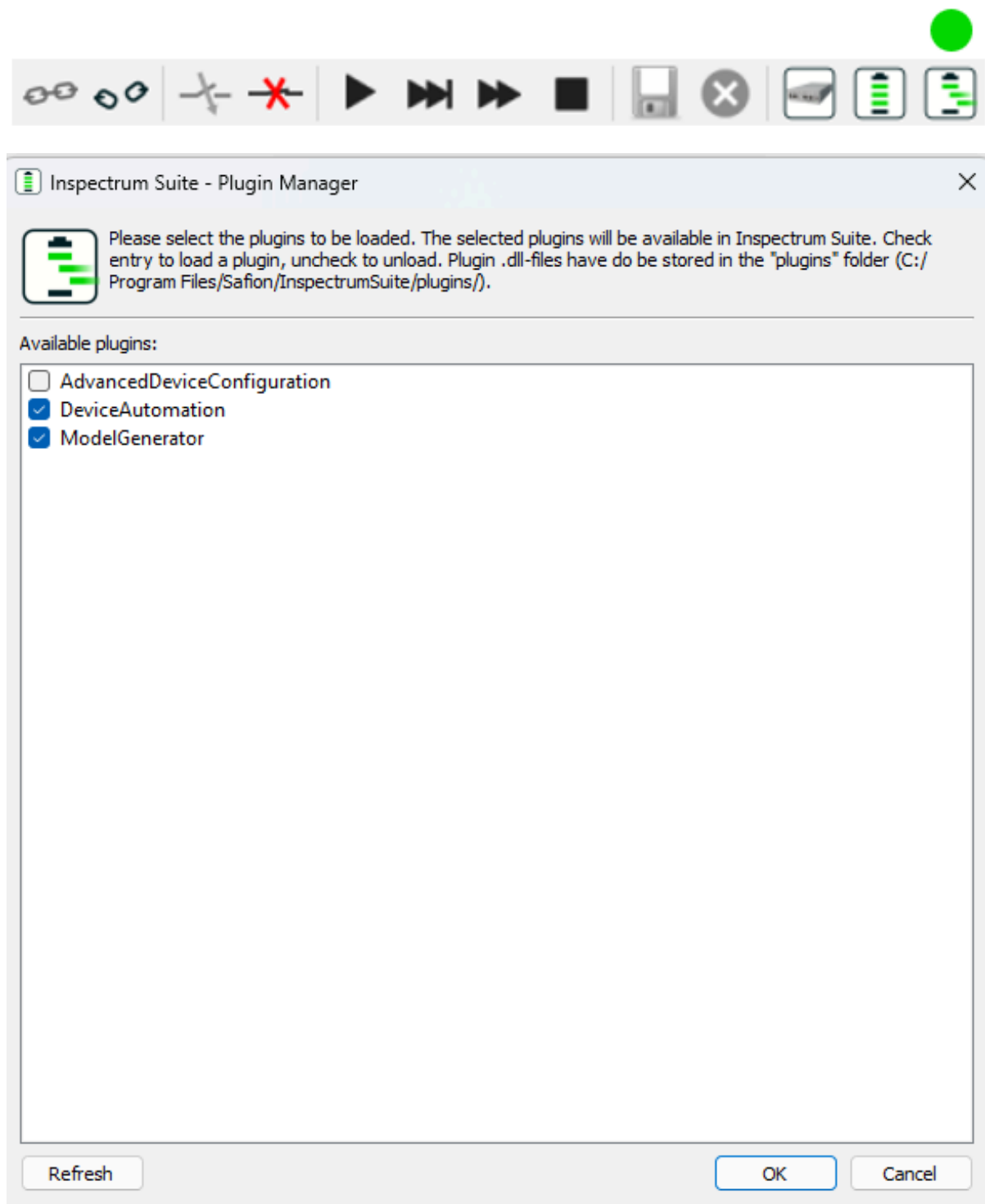


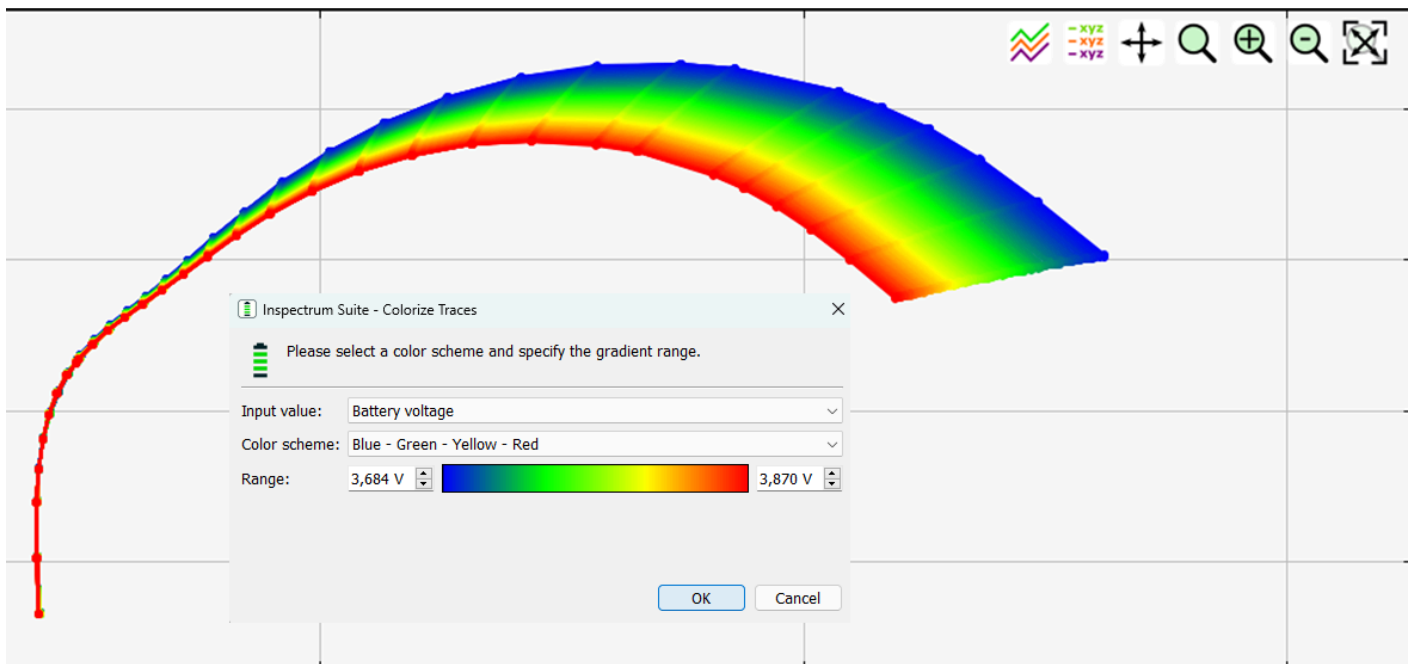
Figure 52: Plugin Manager.

15 Changelog

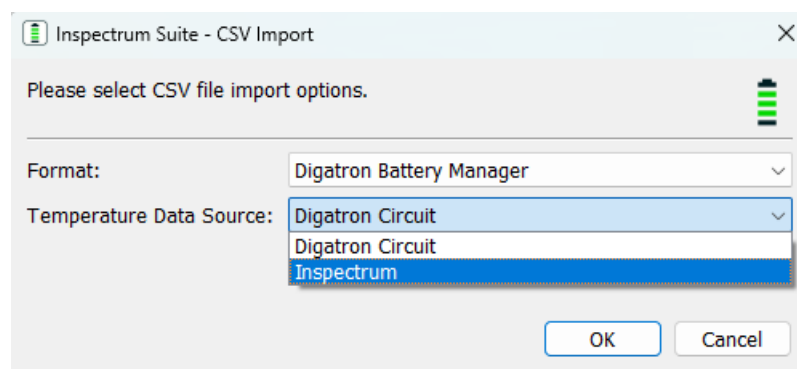
15.1 Version 2.6 (April 2024)

15.1.1 New features

- New HTML based online documentation system.
- Ability to provide displayed measurement data with a color gradient based on temperature or cell voltage:



- The HTML plot export function now has an option to keep the coloring as in InspectrumSuite, which also allows exporting the aforementioned color gradient plots.
- Ability to Import *Digatron Battery Manager 4 (BM4)* CSV files with embedded Inspectrum measurement results.



- New function for spectrum compensation: A reference spectrum can be defined which is subtracted from the actual measurement data. This allows either differential measurements (Delta-EIS) to be carried out, or the influences of cell holders and cabling to be eliminated by means of a short-circuit measurement.

Spectrum Compensation:

Apply to:

Compensation Source:

Mode:

Both

Real Part

Imaginary Part

- Drift correction: Starting with firmware version 1.10, the Inspectrum device now supports linear as well as exponential drift correction methods. This allows to greatly reduce measurement artifacts when performing EIS during charge or discharge operations.

Voltage Drift Correction Mode:

Continuous measurement period [s]:

Frequency table:

- New installer framework for installation of Inspectrum.Suite and plugins.
- The workspace explorer widget now has a history of recently used workspace directories.
- The measurement data preview window during sweep measurements now supports zooming and panning.

15.1.2 Bugfixes

- HTML Nyquist plot export: Y-Axis sign is no longer inverted.
- Fix a problem of firmware files, under some circumstances, do not show the correct version numbers.
- When exporting a CSV file, the path of the source file is now correctly suggested as default.

16 Troubleshooting

16.1 Error Codes

The most common error codes that can occur during operation are described here. In most cases, the error is caused by a bad battery connection or a completely disconnected battery. The error code and description will be visible above the measurement tab and the error can be cleared by clicking the *Reset device error* button.

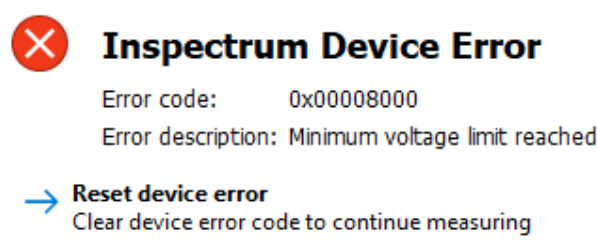


Figure 53: Device Error Code.

If the Inspectrum.C is not able to clear the error, please try powering off and on the device.

Error Codes	Description	Solution
0x00000800	Battery Voltage is not within the supported range of the Inspectrum.C	The battery voltage must be in the range of 0 V to 5 V. Check the battery polarity, voltage, and connection
0x00004000	Current output reached current limit	Reduce the excitation current amplitude or increase the current limit (maximum of 5A)
0x00008000	Battery voltage is below minimum voltage limit	Reduce the lower voltage limit or increase the battery voltage
0x00010000	Battery voltage exceeds maximum voltage limit	Increase the upper voltage limit or decrease the battery voltage
0x00020000	Battery is disconnected	The battery must be connected with the "Connect Battery" button in the software toolbar before starting a measurement
0x00080000	Measurement is running	One or more of the configuration settings have been tried to be modified during a measurement. Stop any running measurements before altering the configuration settings
0x00100000	Multiplexer Error / Multiplexer disconnected	The connection between the Inspectrum.C and the multiplexer has been disturbed. Check the connection and restart the

Error Codes	Description	Solution
		Inspectrum.C
0x00800000	Temperature sensor error	The temperature measurement frontend is malfunctioning. Try a restart of the Inspectrum.C. If the error persists, contact the manufacturer
0x01000000	User Config Error	One or more device configuration values are out of the supported range. Check the parameters and retry to transmit the settings
0x80000000	Emergency shutdown	A fatal error occurred during a measurement (battery disconnected, overvoltage event, overcurrent event), forcing the device to stop the process instantly. Check the battery connection and voltage.

II. Device Automation Plugin

17 Introduction

The Device Automation plugin offers a convenient way to automate complex processes whenever multiple steps need to be executed. Measurements can be configured and triggered when specific conditions are met or in a row of consecutive measurements. The plugin offers a wide range of functionality controlled with a visual programming interface.

Device Automation

Plugin version: 0.6

Automation schedule:

	Instruction	Parameters	Comment	Error action
1	Start charge / discharge process	Target: 2.6 V; Max. current: 1 A; Threshold: 0.97 A (10 s)	-	Stop execution
2	Wait for charge / discharge ...	Timeout: infinite	-	Stop execution
3	Wait	Time: 10:00	-	-
4	Start charge / discharge process	Target: 3.5 V; Max. current: 1 A; Threshold: 0.97 A (10 s)	-	Stop execution
5	Start voltage and current export	File name: <WORKSPACE_PATH>/chargeMeasurement.cs...	-	Stop execution
6	New measurement results ...	Container: results	-	Stop execution
7	Set measurement results contain...	Container: results; Status: enabled	-	Stop execution
8	Label	Name: measureCha	-	-
9	Perform single measurement	-	-	Reset errors an...
10	Export measurement results ...	Container: results; File name: C:/Users/safadm/Desktop/...	-	Stop execution
11	Wait	Time: 01:00	-	-
12	Branch on charge / discharge	Condition: Charge / discharge process running; Target lab	-	Stop execution

Variables:

☒ Show variables

Name	Type	Value
------	------	-------

Start device automation

Execute automation schedule starting with the first instruction

Figure 54: Device Automation: Overview.

18 Installation

The plugin needs to be selected in the plugin manager to start the Device Automation plugin. The **plugin manager** can be found by clicking the respective icon in the toolbar.

Within the Plugin Manager, if the Device Automation plugin is visible in the selection, it can be activated by simply checking the box and applying the changes with the "OK" button.

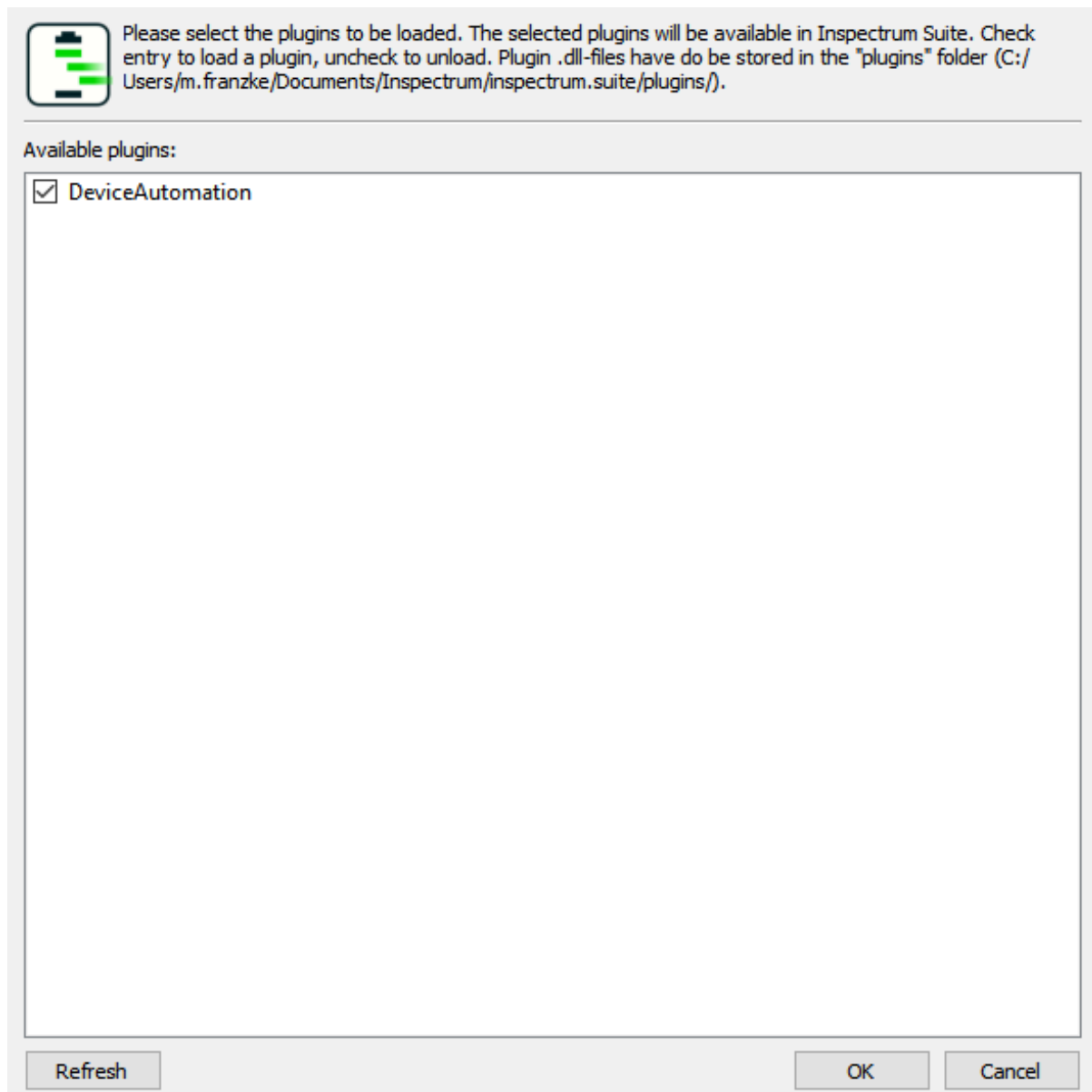


Figure 55: Device Automation: Activating within Plugin Manager.

In case the plugin is not available for selection, it is likely that it needs to be installed. To resolve this, follow the instructions provided by the installation wizard. This wizard will guide you through the installation process step by step. Once the installation is successfully completed, the plugin should become available for selection in your application.

However, if following the installation steps does not yield the expected results, there is an alternative solution. You have the option to manually copy the specific .dll (Dynamic Link Library) file associated with the plugin into the appropriate directory. The path to this directory is displayed in the header of the plugin manager. This method is useful when the automatic installation process encounters issues.

Rest assured, by carefully following the recommended steps, you should be able to make the plugin selectable and fully operational within your system. If you encounter any further difficulties, do not hesitate to seek assistance from our support team. Once the plugin is activated, it will initialize all necessary libraries, and it is ready to be used.

To start the plugin, please select the "Device Automation" plugin tab below the Inspectrum.Suite main area.

This will transform the main area to the plugin view.

19 Overview

This section provides a comprehensive overview and explanation of the interface, outlining its key functionalities and how to effectively navigate and utilize its features.

The Device Automation plugin view is made up of four main selection areas.

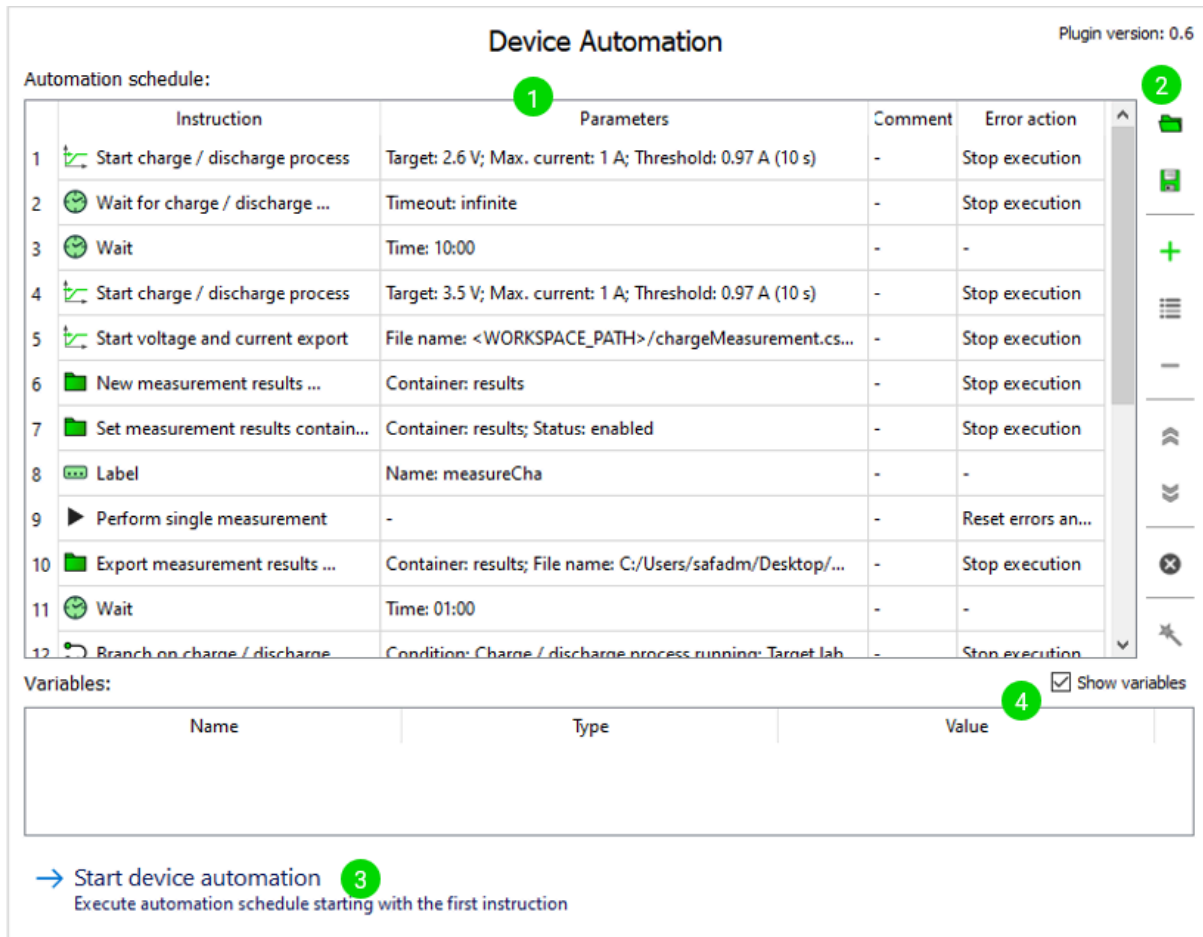


Figure 56: Device Automation: Overview.

- Automation schedule:** The Automation schedule is a list of instructions. Here all steps of the current program are displayed. Each step contains an instruction, the corresponding parameters and if specified a comment and the error action.
- Control Panel:** The control panel allows users to work with the automation schedule. You can add, remove, or change the order of instructions. You can also save your work, or open a previously made automation.
- Start device automation button:** First checks the automation for any mistakes like missing labels. After that, it begins running the automation.
- Show variables checkbox:** Toggles the display of the variable area on or off.

19.1 Context Menu

The context menu can be accessed by right-clicking on an entry within the automation schedule, offering convenient access to a selection of configuration options. These settings are also adjustable through the control panel.

19.2 Control panel



1. Add instruction.
2. Clear automation schedule.
3. Export automation schedule.
4. Import automation schedule.
5. Insert instruction.
6. Loop wizard.
7. Move selected instruction(s) down.
8. Move selected instruction(s) up.
9. Remove selected instruction(s).

20 Schedule Creation

The first step to start the automation is to create a process schedule. The schedule is created by adding an instruction for every step that needs to be executed.

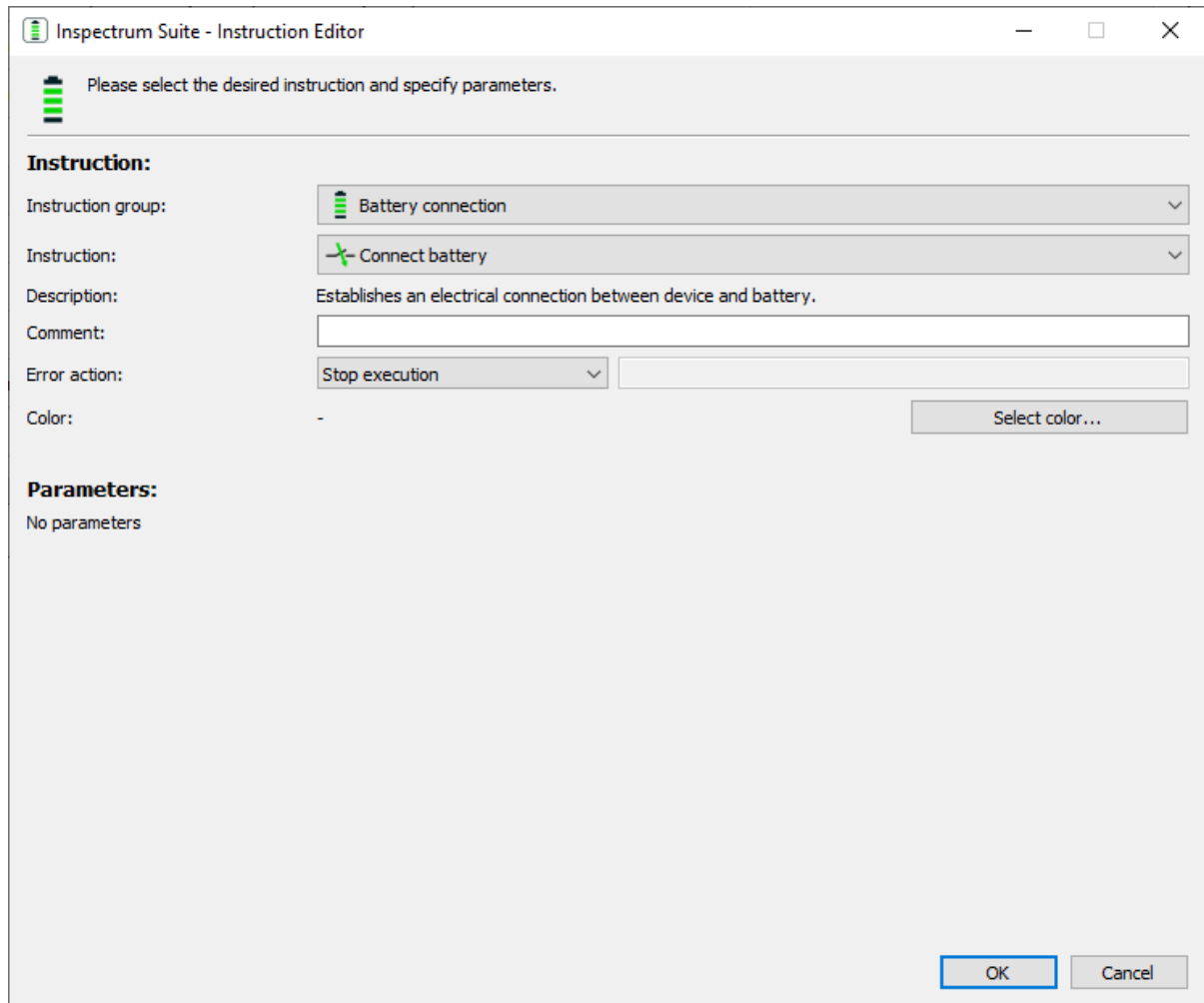


Figure 57: Device Automation: Instruction Editor.

20.1 Instructions

To create schedules, you select the relevant entry in the instruction editor. You can open the instruction editor by clicking the + symbol in the control panel. The initial step is to choose the instruction group. In this group, users are presented with 17 different options to select from.

20.2 Description of the instruction set

Icon	Instruction	Description
	Battery Connection	Connect the battery, disconnect the battery, and change the automatic connection and disconnection status. Before every measurement, the battery must be connected if this is not the case. If the automatic battery connection is enabled, this will be done automatically.
	Battery Monitor	Start and stop battery voltage and current export. When this instruction is active, no other files can be exported. This fact also includes exporting result files.
	CAN Interface	The device automation allows controlling the CAN port. This instruction can be used to react to CAN messages and also send CAN messages to connected devices.
	Charge/Discharge Battery	Handle the charging and discharging process and set the parameters.
	Device Configuration	Import configurations from a file and change numerous device parameters.
	Device Connection	Connect or disconnect to an Inspectrum device, either via USB or network connection.
	Device Error Handling	Branch to a label when the connected device reports an error. You can define separate labels for different kinds of errors or clear the error.
	Flow Control	Branch to a label, if a specific condition was fulfilled. Possible conditions are: - CAN message received, - Charge / discharge process running, - Continuous measurement running, - The device reports an error, - Compare the elapsed time, - Compare measurement results, - Compare variable values. Additionally, it is possible to branch without a condition. Furthermore, it is possible to call a function. A label and a corresponding return instruction define this function. After the return instruction, the automation schedule will resume behind the call instruction. With the end-of-schedule instruction, the automation will be determined.

Icon	Instruction	Description
	Inspectrum Suite	Define export properties like battery name, export comment, and export file or enable/disable automatic export of measurement results.
	Label	Define a label. This instruction can make the automation more-human-readable or as an entry point to branch to / call.
	Measurement	Perform or interrupt measurements. The three options available in this tool are single measurement, single sine sweep measurement, and continuous measurement. Of these three, only the continuous measurement can be interrupted during running automation. The instructions to wait for the measurement stopped can also be chosen if a single sine sweep or a continuous measurement is running.
	Measurement Result Container	Multiple measurement results can be grouped into a result container. These containers are an easy way later to export all measurement results into a single file. Multiple containers can be used simultaneously, and to select which container to save the current data to, it needs to be activated. The containers can also be cleared and deleted.
	Multiplexer	If a multiplexer is connected to the measurement setup, this instruction is used to select the current measurement channel. It is possible to use a variable to set the channel. This way, going through the multiplexer channels in a loop is possible. If channel 0 is selected, this will deselect the multiplexer channel.
	Scripting	This instruction set allows the user to run a Python script. When executing a script, the automation plugin will look for and execute a run() function. Previously defined variables can be passed as parameters; if there is a return value, this can be passed to a variable. The script will still be executed if no run() function is defined, but the return value will always be zero. The working directory of the Python script can be defined. Data in this directory can then be imported into the script. Python scripts can be executed on measurements as well either on a specific or the most recent measurement results container. In both cases, the data will be available within the Python code in a global variable called "measurementResults".

Icon	Instruction	Description
	Timing	Instruction set to work with timestamps and delays. Timestamps use the format: "YYYY/MM/DD -- HH:MM: SS". If a timestamp is initialized with the value "0" it will automatically be set to the current time.
	Variables	Declare variables and set their values.
	Visualization	Create a blank line as a section divider to make the schedule more human-readable.

21 Scripting

21.0.1 Introduction

The Device Automation Plugin allows users to execute pre-existing Python scripts to extend its functionality. These scripts can receive measurement results and perform actions based on this data. This chapter provides detailed instructions on how to execute scripts using the Device Automation Plugin.

21.0.2 Loading Existing Scripts

Before executing a script, it must already exist. Follow these steps to ensure you have the scripts you need:

1. Create or obtain the necessary Python scripts externally.
2. Store these scripts in a location accessible to the Device Automation Plugin.

21.0.3 Testing Python Scripts Prior to Deployment

Before integrating Python scripts into the program, it is crucial to thoroughly test them. This ensures smooth functionality and mitigates potential errors. There are two primary methods to conduct these tests

1. Using an Appropriate Integrated Development Environment (IDE):

A robust integrated development environment provides the ideal space to test Python scripts. Here, developers can write, debug, and execute code while having access to comprehensive tools and resources to thoroughly review the scripts.

2. Execution with the Scripting Plugin:

The Scripting Plugin offers a flexible option for testing Python scripts. It consists of three parts:

- In the left section, measurement values can be inserted and managed.
- The middle section is reserved for the Python code itself.
- The right section displays the output of the Python script.

This plugin not only allows the execution of the scripts but also provides the capability to pass measurement data. This enables developers to simulate realistic scenarios and ensure that the scripts function reliably under various conditions.

The choice between these two methods depends on the specific requirements and preferences of the developer. Regardless of the chosen method, it is important to allocate a thorough testing phase to ensure that the Python scripts can be seamlessly integrated into the overall program.

21.1 Executing Scripts

21.1.1 Configuring Script Parameters

When executing a script, the automation plugin will look for and execute a `run()` function. Previously defined variables can be passed as parameters;

if there is a return value, this can be passed to a variable. The script will still be executed if no `run()` function is defined, but the return value will always be zero. The working directory of the Python script can be defined. Data in this directory can then be imported into the script.

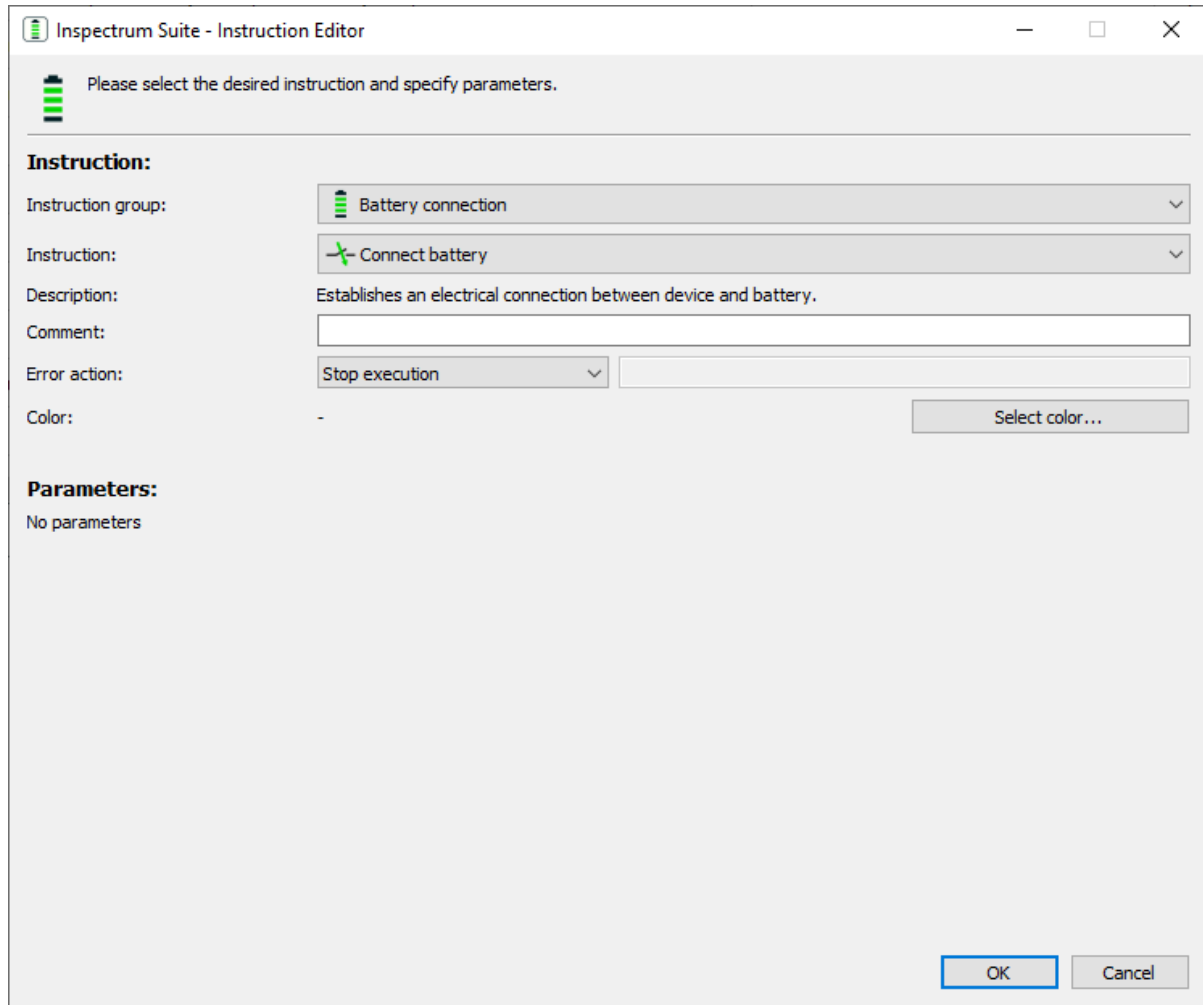


Figure 58: Instruction Editor: Scripting Command.

21.1.2 Using Measurement Results

It is possible to transfer results from EIS measurements to the python scripts, and perform actions based on that data. The results are introduced to python as a global variable called `measurementResults`. The data structure is summarized in this image:

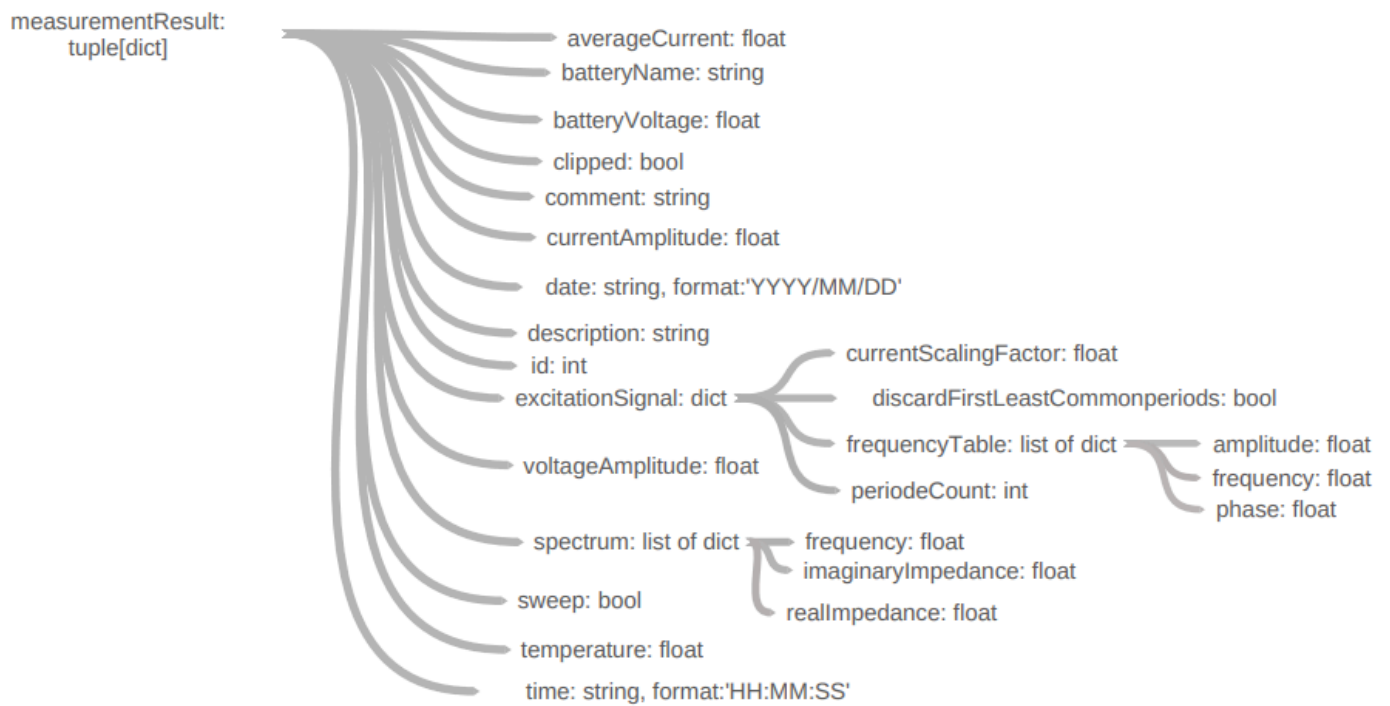


Figure 59: Measurement Result Container: Data Structure.

21.1.3 Initiating Script Execution

To execute a script, follow these steps:

1. Open the Device Automation Plugin interface.
2. Add a new instruction.
3. Select the desired script from the available list.
4. If required, provide the necessary parameters.
5. Click the *Execute* button to run the script.

21.1.4 Troubleshooting

If the automation does not run properly, additional information can be found in the log-file. This is accessible through the plugin preferences. For further information see next section.

22 Preferences

The preferences of this plugin are integrated into the preferences of the Inspectrum.Suite once the plugin is loaded. An additional tab is added for each plugin. Within this tab, all settings of the device automation can be adjusted.

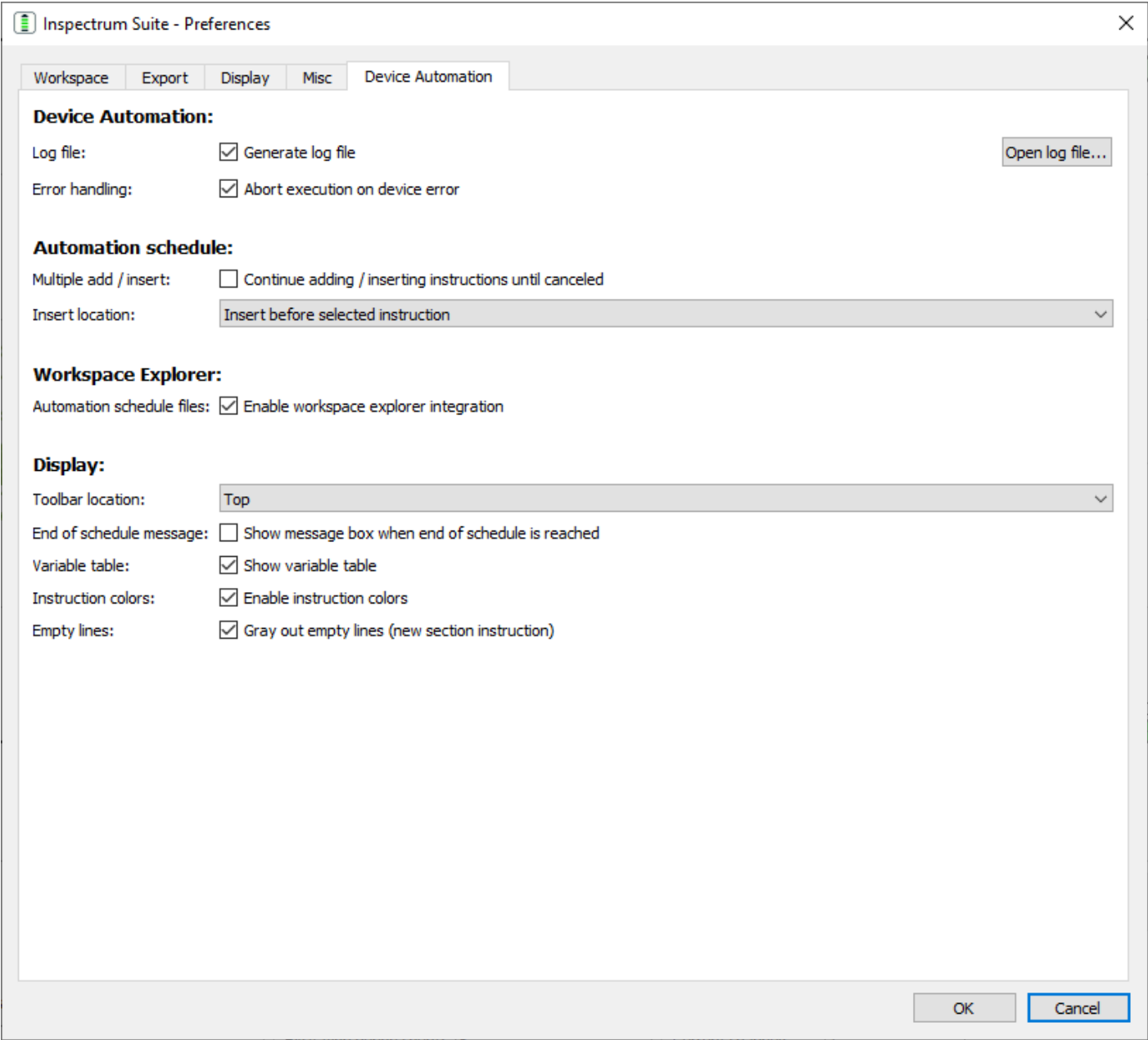


Figure 60: Plugin Preferences.

Item	Description
Log file	Enable this function to allow the plugin to create a log file. From here you have the option to read the log file. Pressing the "Open log file..." button will automatically open the log file in the default text editor.

Item	Description
Error handling	Sets the global behavior of how to handle an error.
Automation schedule	Define how new lines are inserted.
Workspace Explorer	Enable or disable workspace explorer integration. This checkbox will either display or hide the automation schedule files (.ias).
Display	Lets the user choose the location of the toolbar, turn on/off an end-of-schedule message, and how to handle instruction colors and empty lines. Also, the variable view can be enabled and disabled.

III. Model Generator Plugin

23 Overview

The Model Generator plugin offers a full range of battery modelling functions. This can be used for generating simulations and predictions and in control loops to enhance the precision.

The plugin can create a grey-box model based on an equivalent circuit structure. A couple of pre-defined model structures for battery parameter modelling are suggested by the plugin. Additionally, it supports user defined model structures that can be input as a string. This procedure is described in chapter "Model Structure".

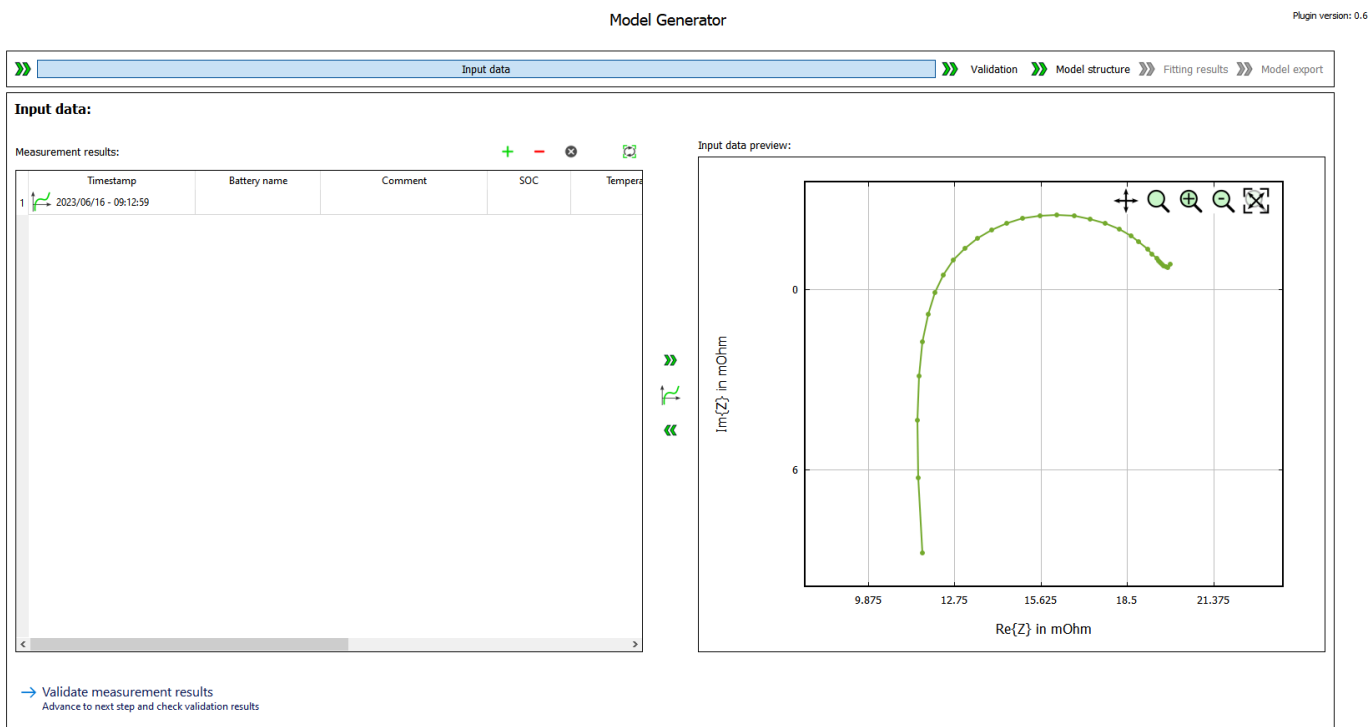


Figure 61: Model Generator: Overview.

24 Installation

The plugin needs to be selected in the plugin manager to start the Device Automation plugin. The **plugin manager** can be found by clicking the respective icon in the toolbar.

Within the Plugin Manager, if the Model Generator plugin is visible in the selection, it can be activated by simply checking the box and applying the changes with the "OK" button.

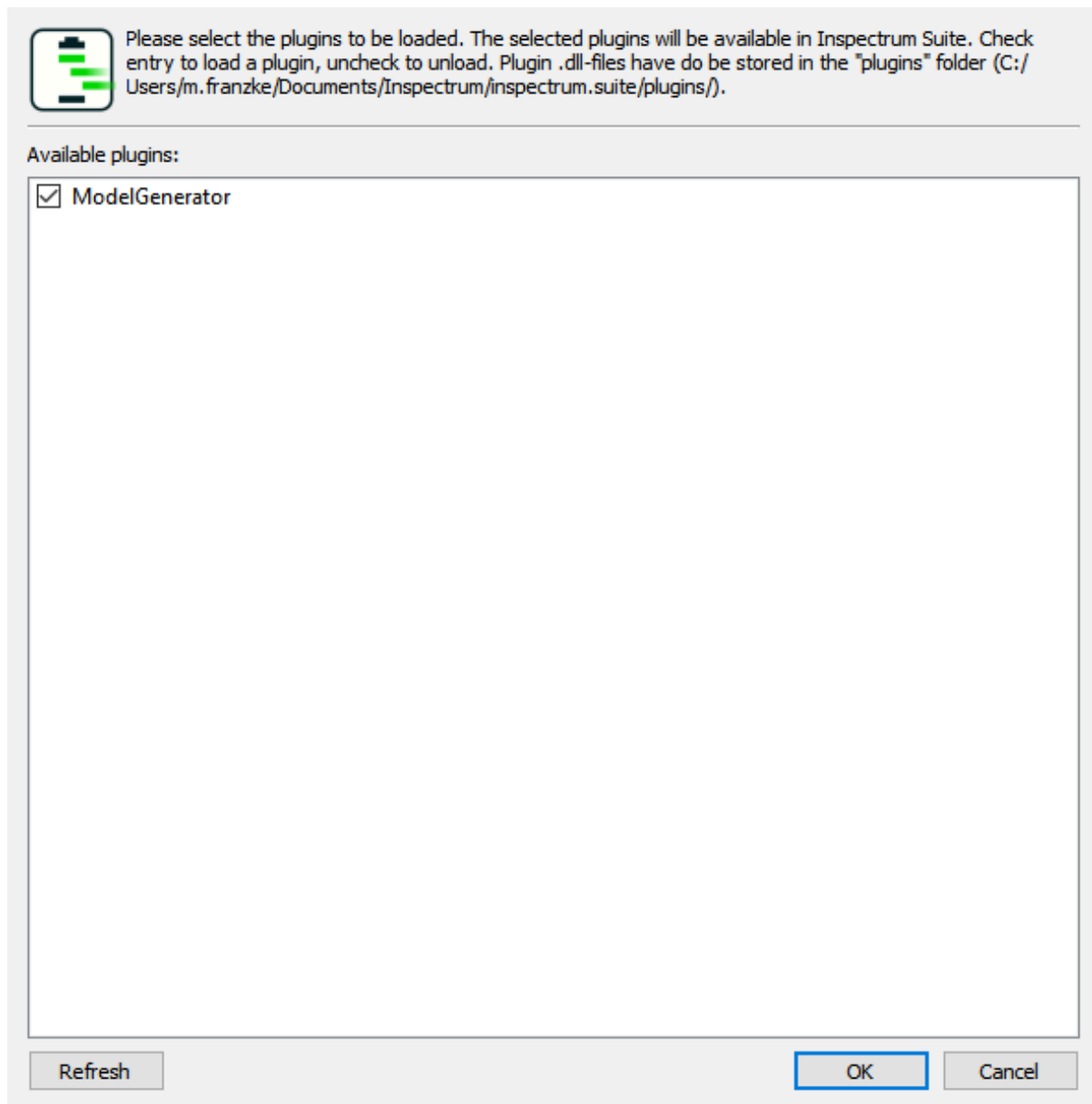


Figure 62: Model Generator: Activating within Plugin Manager.

In case the plugin is not available for selection, it is likely that it needs to be installed. To resolve this, follow the instructions provided by the installation wizard. This wizard will guide you through the installation process step by step. Once the installation is successfully completed, the plugin should become available for selection in your application.

However, if following the installation steps does not yield the expected results, there is an alternative solution. You have the option to manually copy the specific .dll (Dynamic Link Library) file associated with the plugin into the appropriate directory. The path to this directory is displayed in the header of the plugin manager. This method is useful when the automatic installation process encounters issues.

Rest assured, by carefully following the recommended steps, you should be able to make the plugin selectable and fully operational within your system. If you encounter any further difficulties, do not hesitate to seek assistance from our support team. Once the plugin is activated, it will initialize all necessary libraries, and it is ready to be used.

To start the plugin, please select the "Model Generator" plugin tab below the Inspectrum.Suite main area.

This will transform the main area to the plugin view.

25 Model Generation

The model-generation process is divided into five main steps. The current step in the process is highlighted in the Progress Bar. These steps are described in this chapter.

25.1 Input Data

The input data view page lets the user select the data used for the fitting process.

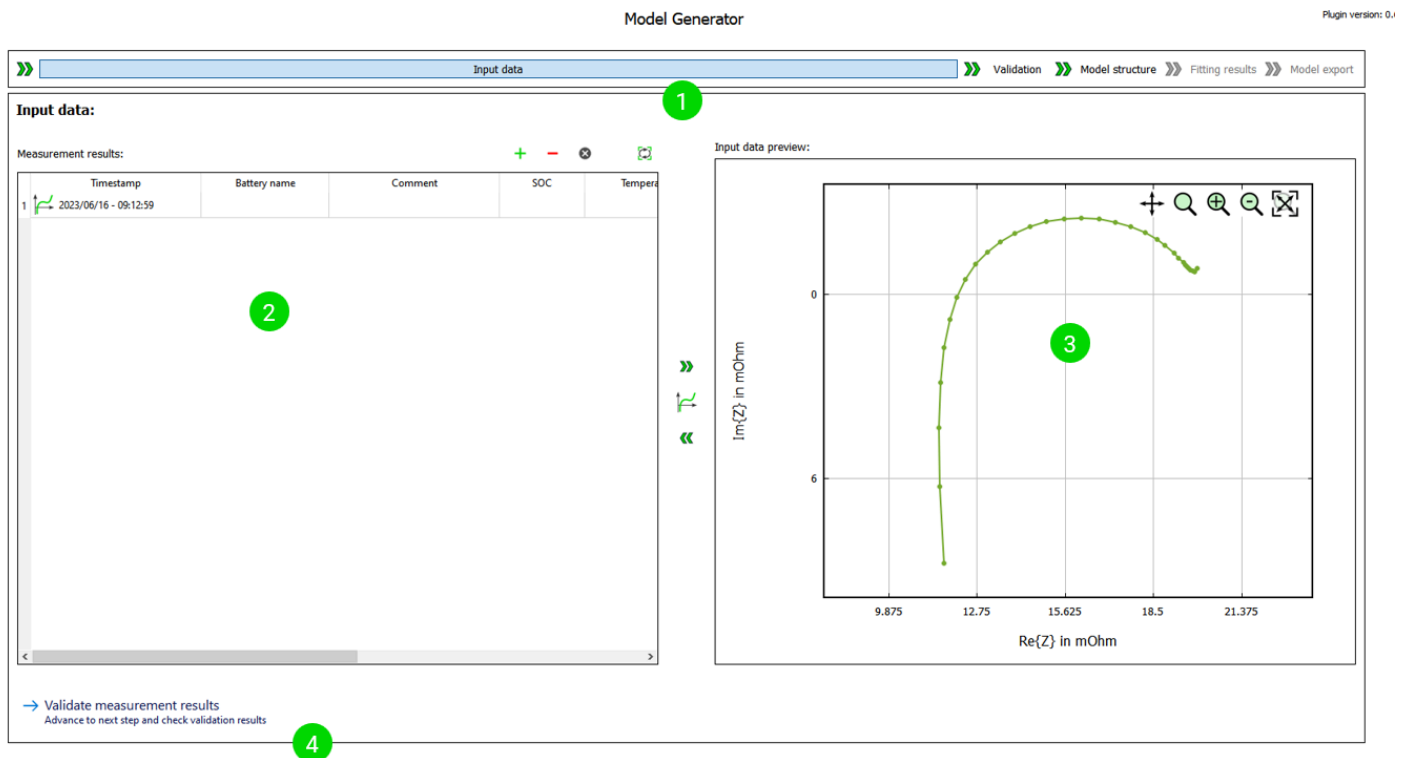


Figure 63: Model Generator: Overview.

1. Progress Bar: Displays the current step and number of total steps
2. Input Area: Shows data used for calculation
3. Plot Area: Shows Nyquist plot of input data and calculated model
4. Next Button: Brings the user to the next step

Measurement results:



	Timestamp	Battery name	Comment	SOC	Temperature in °C	Average group	Group / result name
1	 2023/06/16 - 09:12:59			0.5	25	Group 1	SOC_0.5_TEMP_25
2	 2023/06/16 - 09:14:30			0.5	25	Group 1	SOC_0.5_TEMP_25
3	 2023/06/16 - 09:03:23			0.2	25	-	2023/06/16 - 09:03:23
4	 2023/06/16 - 09:08:36			0.2	45	-	2023/06/16 - 09:08:36
5	 2023/06/16 - 09:05:21			0.8	45	-	2023/06/16 - 09:05:21

Figure 64: Close view of the input area.

1. Add measurement result(s)
2. Remove measurement result
3. Clear measurement results
4. Generate average groups

Selecting the measured data for the model generation process is either done by pressing the import button and using the file viewer dialog or by simply dragging the desired data files to the input area in the main page of the plugin. It is possible to add multiple measurements of the same cell or from different cells. To differentiate between the cells later in the process unique cell names must be defined. This can be done by double-clicking the battery name field of the input data table and assigning a name to the selected battery cell data.

Similarly, the State of charge and temperature of the measured battery cell data can be adjusted if not already correct.

25.1.1 Average Groups:

If there are more than one measurement of the same cell and state, average groups can be used to combine the data. Further calculation will utilize the average values of the spectra of all measurement within this group.

Cells with matching SOC and Temperature will automatically be put in the same Average group if this option is activated in the preferences [4]. If the SOC and Temperature are adjusted by the user, these groups can be regenerated by clicking the button *Generate average groups* (4). They can also be manually assigned by utilizing the dropdowns.

25.2 Validation

Once all data is imported into the plugin, the next step is to validate this data. This step is necessary to ensure that the measured data is plausible so that the plugin will then generate correct results. This means that the spectra need to be checked for causality, linearity and stability.

To check the validity, the spectrum is compared to a generated model, consisting of a number of RC elements in series with a resistor and optionally a capacitance, fit to the data. The premise is since this generated spectrum will be compliant to the conditions of causality, linearity, and stability, so will the measured spectrum if the fit is good enough. The maximum deviation of the two spectra is then used to determine the validity. Usually values of more than 1 % deviation indicate an invalid spectrum and the data should not be used for model generation. This will be indicated by a red circle in the measurement results table.

The validation can be adjusted with the validation options, but best results are generated with the default settings.

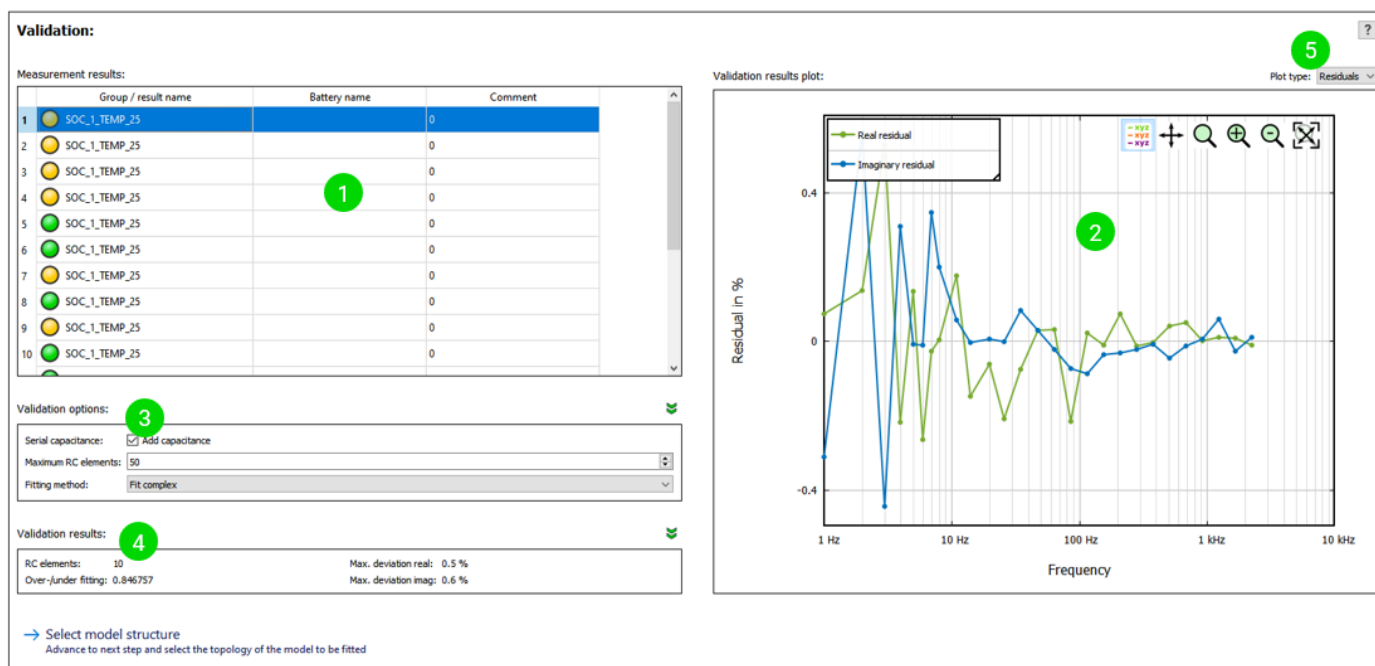


Figure 65: Data validation.

1. Measurement results
2. Validation result plot
3. Validation options
4. Validation results
5. Plot type selector

25.3 Model Structure

After selecting the input data and validation step, the next step is to select the model for parametrization.

Model structure:

Model settings:

Model presets: 1x R, 1x RC, 1x OCV

Model string: R0-p(R1,C1) ?

Model schematic:

→ Generate model

Advance to next step and review fitting results

Figure 66: Model structure selection.

1. Model presets
2. Model string
3. Model schematic

This step can be found in the Model Structure section of the plugin. The plugin provides some predefined models that can be chosen from the model presets dropdown.

For battery model parametrization usually a simple RC-Model or a ZARC-Model is used. If a different model is needed, it is possible to create custom structures as well. For custom structures the available elements are:

Symbol	Description
C	defines a capacitor
CPE	defines a constant phase element
G	defines a Gerischer Element as $Z = \frac{R_G}{\sqrt{1 + j2\pi f t_G}}$
Gs	defines a Gerischer Element as $Z = \frac{R_G}{\sqrt{1 + j2\pi f t_G} \tanh(\phi \sqrt{1 + j2\pi f t_G})}$

Symbol	Description
K	An RC element for use in lin-KK model
L	defines an inductor
La	defines a modified inductance as $Z = L \times (j2\pi f)^\alpha$
R	defines a resistor
W	defines a semi-infinite Warburg element
Wo	defines an open (finite-space) Warburg element
Ws	defines a short (finite-length) Warburg element
ZARC	defines a parallel connection between a constant phase element and an ohmic resistor

Connection of multiple circuit elements

- **p(a,b)** defines a parallel circuit of element a and b
- **a-b** defines a series circuit of element a and b.

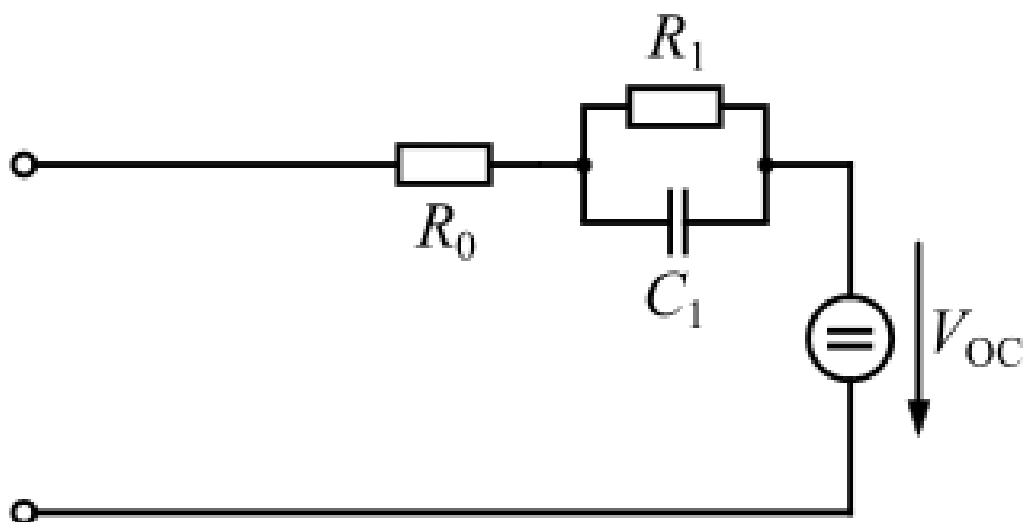


Figure 67: Example of a circuit model $R_0-(R_1,C_1)$ creates one resistor in series with a parallel connection of a resistor and a capacitor.

25.3.1 Fitting Results

The fitting process can be started by pressing the "Fitting results" button. Depending on the complexity of the model and the number of models, this process might take a while to finish. Once completed, the fitting results will be displayed in the Nyquist plot together with the measured data. The fitted parameter values are additionally presented in a table below the input data.

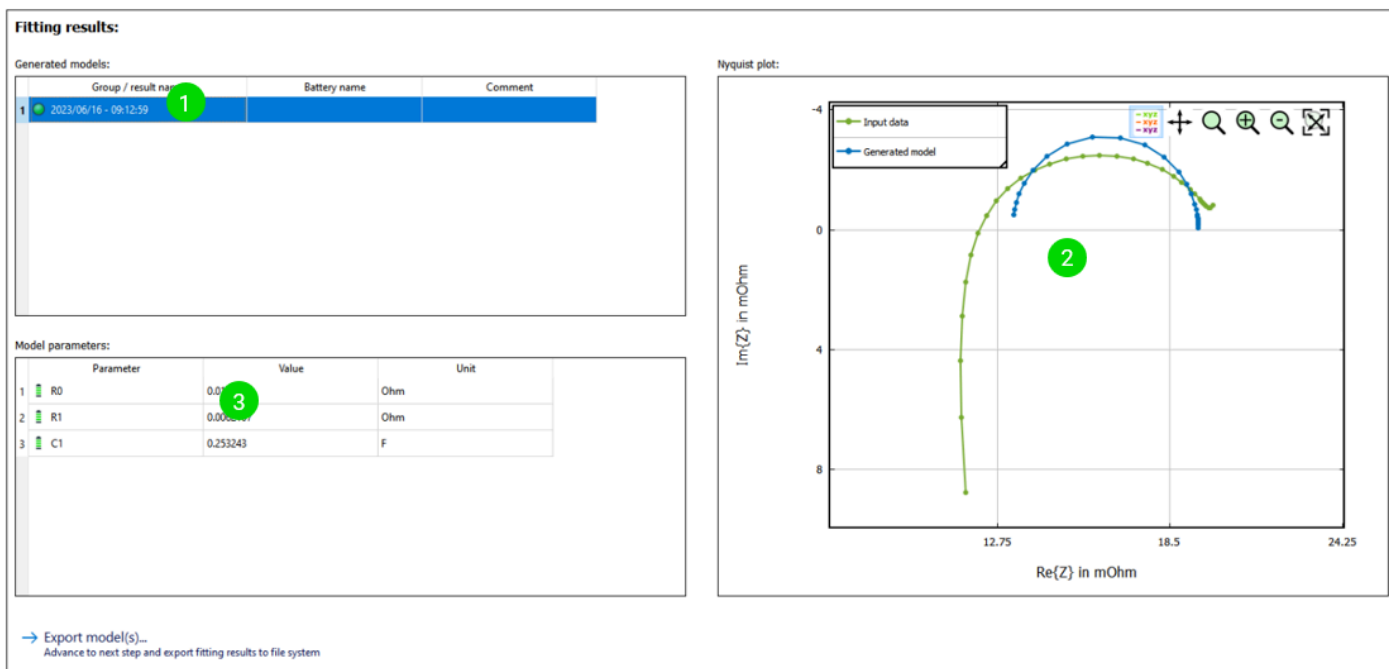


Figure 68: Model fitting results.

1. Generated Models
2. Nyquist plot
3. Model parameters

25.3.2 Parameter Adjustment

After calculating a fitted model all parameter values are displayed in the model parameter table. In rare cases the plugin may not find an optimal solution. Therefore the user is able to adjust all calculated parameters manually. Double-clicking the parameter will open the parameter adjustment view. Therein the value can be changed by moving the slider or by using the value input. The Nyquist plot is automatically updated, when releasing the slider or pressing the "Apply" button if this function is activated in the [preferences](#).

Parameter: R0

Current value: 0.00217449

New value:

0 Ohm 0.00434897 Ohm

Figure 69: Model parameter adjustment

Once the parameter has been changed, an arrow in the value field of the parameter table will indicate the change. If this arrow is pressed, the parameter is reverted to its initial value.

25.4 Exporting Results

When the parameter adjustment process is finished, the resulting models can be exported as an external file that can be loaded, viewed, and changed at a later stage. To start the export process, the *Export model(s)...* button must be pressed. In the window that opens, either a single model or all calculated models can be exported. The exported data will be stored in an ".imf" file format that can store multiple data models and be imported into the Inspectrum.Suite software.

25.5 Importing Results

Previously exported model results can be imported into the Model Generator plugin to compare, adjust, and evaluate the results. This can be done by clearing the plugin input data and then double-clicking the exported ".imf" model generator file. From this point it can be handled just like a newly calculated model.

26 Preferences

The preferences of this plugin are integrated into the preferences of the Inspectrum.Suite once the plugin is loaded. An additional tab is added for each plugin. Within this tab, all settings of the device automation can be adjusted.

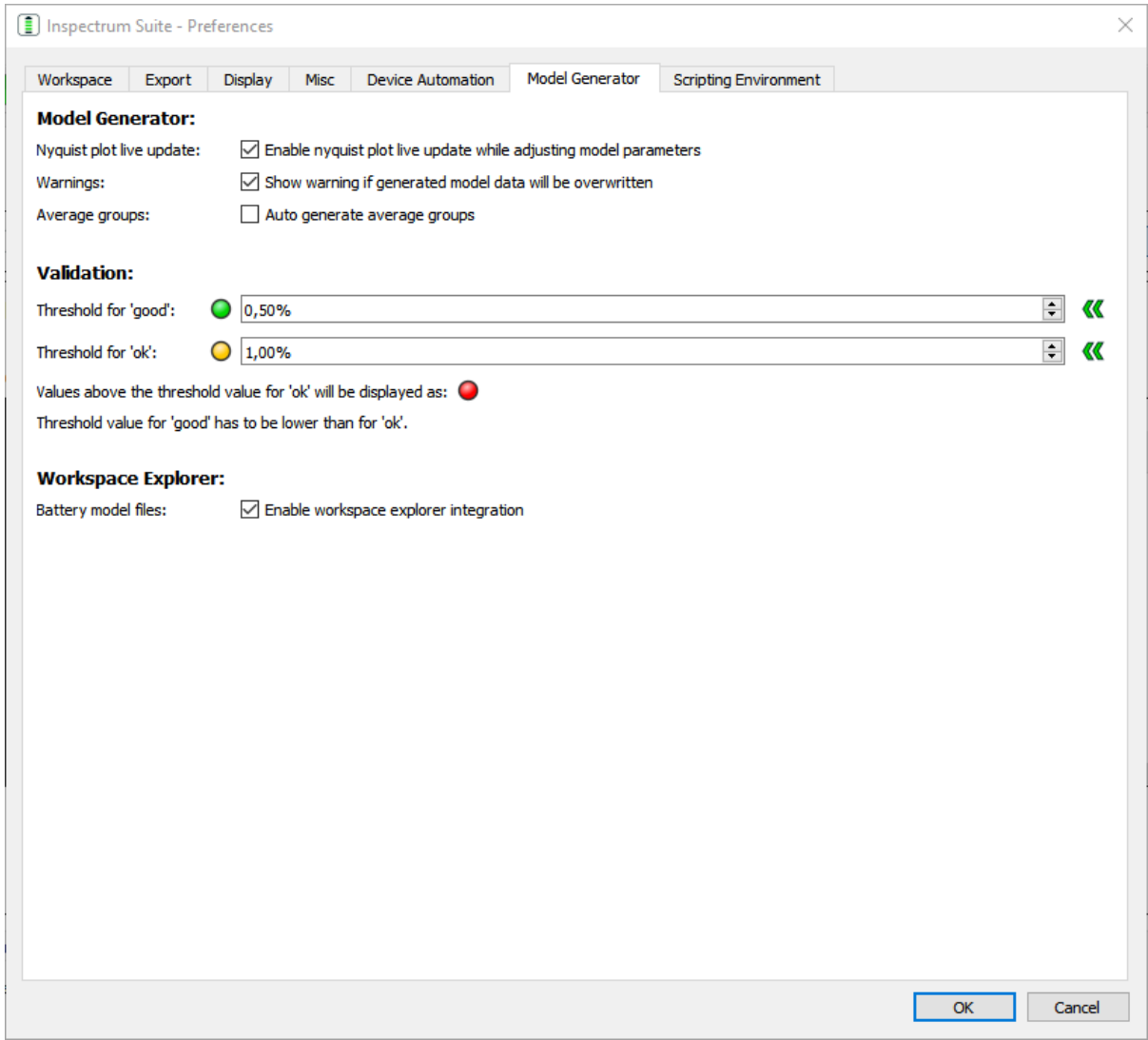


Figure 70: Model Generator preferences

Item	Description
Nyquist plot live update	Enable / Disable the live update function when model parameters are changed.
Warnings	Enable / Disable warnings when generated model data will be overwritten
Average groups	Enable / Disable the auto generation of average groups when new data is imported.

Item	Description
Validation	Set the thresholds for the validity indicators. These can be reset to the default values with the green arrows.
Workspace Explorer	Enable / Disable workspace explorer integration.